

TRIED AND TWEAKED:

Activities to re-energize
peer learning sessions



The Peer-Assisted Learning (PAL) Program
University of Minnesota – Twin Cities

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*University of Minnesota
Peer-Assisted Learning Program*

Table of Contents

Foreword.....	iii
Introduction	ivii
Acknowledgments.....	ix

SECTION 1: Applying Study Skills	1
1.1 Developing a Thesis	2
1.2 Note-taking and Active Reading.....	3
1.3 Organizing Textbook Material	4
1.4 Focusing on Concepts with Note Cards.....	5
1.5 Note-taking in Lecture.....	6
1.6 Note-taking in Lecture – 2.0.....	7
1.7 Partial Outlines with Textbooks.....	8
1.8 Venn Diagrams for Concepts.....	9
1.9 Organizing with Timelines.....	10
1.10 Peer Reviews of Essays.....	11
➤ Handout: Writing a Good Thesis.....	12
1.11 Post-Exam Reflections	13
➤ Handout: Post-Exam Reflection	14

SECTION 2: Grasping Concepts	15
2.1 Defining Terms.....	16
2.2 Grasping with Graphs	17
2.3 Making Connections	18
2.4 Using Outside Readings.....	19
2.5 Learning in 3-D	20
2.6 Making Space for Media	21

SECTION 3: Promoting Participation.....	23
3.1 Quiz Quiz Trade	24
3.2 Rotating Worksheets.....	25
3.3 Unprepared!	26
3.4 Warming Up for Discussion.....	27
3.5 Flexible Grouping.....	28
3.6 Student-Led Discussion.....	29

3.7 FlexibleTopics.....	30
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SECTION 4: Solidifying Material	31
4.1 Go with the Flow.....	32
4.2 Exploring Themes	33
4.3 Brainstorming Themes	34
4.4 Journey to Physiology Paradise	35
4.5 Engaging with Real Life Data	37

SECTION 5: Sharpening Problem-Solving Skills.....	39
5.1 Clockwork Math.....	40
5.2 Challenge by Choice	41
5.3 PowerPoint: Is it worth it?	42
5.4 Tutor-Teacher Pairs	43
5.5 Problem-Solving at the Board.....	44

SECTION 6: Reviewing for Exams.....	45
6.1 Lecture Size Sessions.....	46
6.2 Multiple-Choice Madness.....	47
6.3 Reviewing with Sticky Notes.....	48
6.4 Who Makes the Study Guide?.....	49
6.5 Station to Station	50

SECTION 7: Game-Based Reviews.....	51
7.1 Chemistry Bingo	52
7.2 Bingo with Buildings.....	53
7.3 Dueling Teams	54
7.4 Capture the Flag.....	55
7.5 Group Relay	57
7.6 "X"ed out in Tic-Tac-Toe	58
7.7 Jeopardy	59
7.8 Scratch Off with IF-ATs.....	61
7.9 Musical Quizzes	62

7.10 Circle of Fortune.....	63
7.11 Pick Your Present	65
7.12 Quiz Your Peers	66
7.13 Who Wants to Be a Millionaire?	67

SECTION 8: Learning Terms and Vocabulary71

8.1 Blind Key Terms.....	72
8.2 In Their Own Words.....	73
8.3 Matching Sentences.....	74
8.4 Post-It® Vocabulary.....	75
8.5 Vocabulary Bingo	76
8.6 Vocabulary Ice-Breaker.....	77
8.7 Vocabulary Notecards	78

SECTION 9: On the Shorter Side 79

9.1 Warm-Ups.....	80
Lightning.....	80
Model: How to present answers.....	80
Grouping/Problem-Solving Combo	80
Self-Select Grouping.....	80
Reverse Engineering.....	80
9.2 Five-Minute Flares	81
Right or Wrong.....	81
Match the Graph	82
Learning Objective Matrix.....	82

9.3 Post-Exam Self Evaluation.....	83
➤ Handout: Post-Exam Self Evaluation.....	84
9.4 Flippy Pages	85
9.5 Ultimate Exam Page.....	86

SECTION 10: Why Won't You Tell Us!..... 87

10.1 Ask Your Peers First	88
10.2 Confidence is the Issue	89
10.3 No Problem Goes Unsolved	90
10.4 Step by Step	91
10.5 Students as Guides.....	92
10.6 Value of the Struggle.....	93

APPENDIX 95

PAL Session Plan - sample.....	95
Activity Reflection - sample.....	96
➤ Session Planning Form	97
➤ Activity Reflection Form	98

Endnotes:103

INDEX.....104

Table of Contents (cont.)

Foreword

For readers familiar with our first volume, *“Two (or more) Heads are Better than One: Adventures in leading group learning – a facilitator storybook”*, you already know that the Peer-Assisted Learning (PAL) program at the University of Minnesota is committed to sharing the classroom experiences of our undergraduate facilitators. The stories in that book were meant to stimulate discussion of interpersonal practices in handling situations and relationships relevant to a broad spectrum of peer learning programs.

In this second volume, *“Tried and Tweaked: Activities to re-energize peer-learning sessions”*, our undergraduates have assembled a collection of activities they designed and conducted in their weekly sessions. Many of the activities developed from an emphasis on two particular principles (there are eight) that guide our peer-learning sessions: modeling productive learning behaviors, and engaging students with each other¹.

Justin’s introduction provides a modest description of the thoughtful discussions that take place in the weekly team meetings, where facilitators share challenges they are having in planning and conducting their sessions. I too attend these meetings – organized by sciences, math, and humanities – and can vouch that all the leaders and facilitators are guided by the same goal: to conduct engaging and productive sessions that increase confidence, persistence, and deeper understanding of the content. The spirit in which session activities are designed is one of creating an informal, social, and welcoming environment. Facilitators encourage each other to take risks and offer support and advice to help their teammates to be successful.

Our PAL facilitators are further motivated to develop a committed following of attendees throughout the semester and this collection is a testimony to the enthusiasm and courage so many leaders bring to their PAL sessions. Even with careful planning, they often are unsure how things will go, especially when trying something new. Bolstered by good training, a can-do attitude, and the support of their team, the PAL facilitators at Minnesota are a curious and enterprising group; they try, they tell, and they tweak. While no narrative can ever really replace the actual experience of that “first time” playing a game of *“Chemistry Bingo”* (7.1) for a review session, or guiding students to create an *“Ultimate Exam Page”* (9.5), this collection offers inspiration and tips to session leaders who are looking to spice things up. Use them as you see fit; how you do so is of interest to us and we welcome your comments. Thank you for reading and be in touch.

Mary Lilly
Peer-Assisted Learning Program Director
University of Minnesota

Introduction

During my undergraduate studies at the University of Minnesota, I benefitted from several invaluable experiences, but none were more rewarding than the three years I spent facilitating peer-assisted learning (PAL) sessions for students taking chemistry. For two of those years, I also served as the Sciences Team Leader of my six PAL peers. My responsibility was to help my teammates build on their strengths, improve their skills, and become more confident. I did this by asking them to take time during our weekly team meetings to reflect on their classroom experiences. My goal was to inspire the facilitators to use the challenges they faced in the classroom as opportunities to learn and master their craft.

The PAL program at our university provides training and resources to help facilitators design fun and productive activities, but often uncontrollable variables in our sessions can (and do) complicate and derail the best of plans. I discovered that if our team didn't consistently reflect on our session plans, facilitators wouldn't notice if students had improved their study skills and their understanding of the concepts. As a result, facilitators would miss opportunities to adjust their activities to suit their specific classroom environment. We talked at length about these issues at our meetings, critiquing and tweaking activities based on our goals for the session. A key component was a discussion of the successes and struggles of the previous week's PAL sessions – what worked and what didn't. As team leader, I worked hard to create a level of trust so that facilitators would be honest in their recounts; we all liked sharing our notable successes, but it was the “not so glamorous” stories that provided the most fruitful discussions. Through “full disclosure” on everyone's part, problems were discussed openly, good ideas flourished, and we all saw real improvement in our leadership and in the productivity of our PAL sessions. Reflecting on our own performance, and learning to be intentional in our plans for every session, became so important to us that we also held discussions on Moodle so our ideas and remedies would be available to future facilitators.

Last spring, University of Minnesota's PAL Program Coordinator Mary Lilly proposed an idea to all the team leaders: creating a book of activities that facilitators could reference while planning their sessions, knowing that the activity had actually been tried and evaluated. We found the Moodle forum to be a convenient way to archive each facilitator's experience of conducting a noteworthy activity – what they did, how it turned out, and recommended adjustments if they were to use it again. What has emerged is an authentic collection of activities that have been tailored to specific courses to achieve specific goals for the session.

Introduction

Planning sessions with *specific goals* is a big part the PAL training that all facilitators participate in each semester – every PAL session needs to have *learning objectives* that describe what the students will be able to do at the end of (and hopefully because of) the session. The template we use (along with an example) can be found in the Appendix. In our team meetings, we discussed being intentional with the students about the goal, and not simply showing up with a random list of homework problems for the students to trudge through. When deciding that goal, facilitators were expected to consider the type of material being learned, the degree to which the students understood the concepts, and the unique group dynamics of their sessions. As we considered how to organize the stories in this book, we chose the same criteria. The title of each section indicates a general goal that guided the facilitators when they designed their activities; these include: *Applying Study Skills*, *Grasping Concepts*, and *Reviewing for Exams*, to name a few.

It should come as no surprise that we discovered that group dynamics vary dramatically from one group to the next; an activity might work great for one, but fail miserably in another. For example, in the story, *Student-Led Discussion* (3.6), Jeanna used PowerPoint to introduce questions and promote discussion. Since the majority of her students were very outgoing and enthusiastic about the subject, this turned out to work very well. On the other hand, in the story *PowerPoint – Is it worth it?* (5.3), Samara struggled to get good participation out of her students because too many of them were distracted. She describes a large group of 20 plus students, many of whom were either uncomfortable with starting discussion, or needed a more stimulating and less passive activity. Just because a group tends to be very quiet does not mean that an activity won't work, it just needs to be delivered in a format that fits the dynamics of the group. We found it was usually easier to adjust our activity than to change the students. Other typical dynamics facilitators encountered were sessions of either too many introverts or too many extroverts; sessions where students were too good of friends and easily got off topic; overwhelmingly anxious students; and groups that either really enjoyed or intensely disliked competition.

Since the story about each activity in this collection arises out of a unique setting, some activities may not fit your specific environment. Our hope is that you will combine your current knowledge as a peer facilitator with the many choices of activities we present in this volume to ultimately structure an engaging and productive session plan that will bolster your confidence, even before entering the classroom! This collection of “tried and tweaked” activities will re-energize not only the session atmosphere, but you as the facilitator as well. We look forward to inspiring new and veteran facilitators alike so they can develop and archive their own collection to pass on.

*Justin Paz
University of Minnesota
B.S. Biochemistry, 2013*

Acknowledgments

It wasn't too long after "Two (or More) Heads" rolled off the press when I began to think about sharing the creative talents that our PAL facilitators employ when planning their weekly sessions. Already familiar and well-practiced in the art of reflecting and storytelling, these students come to our team meetings to report the inventive and engaging activities they have planned, along with honest recollections of the success (or lack of) they have experienced in executing them. Once again I was fraught by the thought that when those facilitators graduate their creativity and experience go out the door with them; would each new group have to reinvent the wheel?

It became clear that a second volume of first-hand narratives was the route to go in an attempt to capture the wisdom and brilliance of such talented students. I was fortunate to have a motivated, organized, and extremely diligent facilitator – Justin Paz – to lead the effort. Justin is a three year veteran of the PAL program, with steadfast enthusiasm for the many benefits of peer-assisted learning sessions, for both students and facilitators. A creative planner and a willing experimenter, he gathered the narratives – by way of our program's Moodle site – edited them, and developed a user-friendly format for the collection. I am so grateful that he has given so generously of his time these past 13 months, even while fully immersed in the rigors of the first year of dental school.

As with the previous volume, I could not have produced the product you hold. In addition to Justin's dedication, many facilitators contributed their stories with a similar dedication to sharing experiences so that others can benefit from both their successes as well as the not-so. Thank you to all who so generously offered such authentic narratives for this project: Aaron, Addison, Alina, Alyssa, Amanda, Amber, Andrew, Beth, Cameron, Deanna, Elaina, Hannah, Jake, Jeanna, Jennifer, Jered, Jessica, Justin, Kaiwen, Katie, Kelley, Lauren, Lia, Matt, Mercedes, Mindy, Rachael, Samantha, Samara, Sara, Summer, Tali, Tara, Tim, and Trent.

Thanks to the continued trust in our vision by Lynell Williams who, as Director of the SMART Learning Commons, oversees various academic support programs at the University of Minnesota-Twin Cities.

My continued appreciation and gratitude goes to Dr. David Arendale for providing the peer-learning community with his scholarship on learning assistance and, in particular, the guiding principles for PAL group facilitators; his support and enthusiasm for this project have been unwavering.

*Mary Lilly
PAL Program Director
University of Minnesota*

SECTION 1: Applying Study Skills

The purpose of peer-assisted learning goes far beyond simply facilitating students' learning of the material. PAL is an opportunity for students to discover and practice a variety of study methods specifically suited to the course content as well as to learn how to identify their weaknesses and overcome them while preparing for the next exam. The activities in this section are designed to help students to practice using various study techniques to help them decide what works best for them.

College can be very demanding and it requires adjustments in students' study methods, often course by course. Activities in this chapter open the door to methods the student may not be aware of. By modeling how these alternatives work directly with the course content, the facilitator shows students how to apply them. From creating Venn Diagrams, to organizing notes, or doing post-exam reflections, these activities are great for both the beginning of the semester and when students are looking for different ways of studying because they struggle to understand and retain the material for exams and writing papers.

~Justin

Improving students' [study skills] appears to involve raising their awareness of what they are doing, increasing their repertoire so that they can choose to do different things when it seems appropriate and tuning them in to task demands so that they can recognise what is required.

– Graham Gibbs, Raising awareness of best-practice pedagogy

Applying Study Skills

1.1 Developing a Thesis

Supplies/Prep: Post-it® notes with assigned roles for each group (Coordinator, Recorder, Critic, and Speaker); knowledge of what makes a good thesis statement.

Students in the English literature course I facilitate for sometimes struggle with developing a good thesis for their writing assignments. In order to help them with this, I facilitated an activity that would help them create good discussion questions, which could then be used to set up a thesis. At the beginning of the session, we discussed as a group what made a good thesis.

Next, I put the students into groups of four to create discussion questions based on a recently assigned text. After each group came up with at least two discussion questions, I had students trade their questions with another group. The students were given time to think of an answer and then briefly present it to the class. Following the presentations, students had to describe a thesis topic they could create based on any of the discussion questions presented today. To confirm students' comprehension of the day's discussion, I had the students turn in their thesis statements at the end; I was pleasantly surprised with their work.

~Amanda

Subject Area:

- Humanities
- Social Sciences

Grouping:

- Small groups
- Large discussion

Notes:

To make sure everyone participated in some way, I used Post-it® notes to assign roles (Recorder, Critic, Speaker, and Coordinator) to each member of the group.

1.2 Note-taking and Active Reading

Supplies/Prep: Copies of two recently assigned readings.
Prepare several questions for each reading - to be shown on the screen, projector or given as handouts.

I used an activity to practice note-taking skills and active reading skills. In preparation, I selected two readings (each short enough to read in 15 minutes) that the students had been recently assigned in lecture; I then generated questions from the readings.

I asked students to individually read the first article and take notes in a separate notebook. I suggested they record the important themes, events, and issues, as well as identify confusing vocabulary. After they finished reading the text (15 minutes), I removed the article so they only had their notes. I showed students the questions and asked them to first answer the questions and then reflect on the quality of their note-taking skills. They had a difficult time answering the questions the first time around.

At this point I encouraged them to use each other as a resource and see how their peers took notes, observing different note-taking methods. As a large group, we discussed how to identify what is important in the article. I asked them to share their methods of taking notes and describe what aspects were most and least helpful. I repeated this entire process with the second reading. The students improved their ability to identify important themes and were better able to predict the questions. They also adjusted the way they took their notes, using what they had learned from their peers and their own note-taking the first time around. It was exciting to see the progress they made simply by tweaking some of the skills they already possessed.

I find that the value of this activity is that it gives students concentrated time to recognize their weaknesses and begin to make improvements. With the second set of questions they showed a lot more confidence.

~ Mercedes

Subject Area:

- Humanities
- Social Sciences

Grouping:

- Small groups
- Large discussion

Notes:

This activity works well early in the semester so students can assess and improve their note-taking skills, especially in a class based heavily on readings.

Use questions already prepared in the textbook that go along with the readings when writing the questions for the session. It saves time and highlights what the text identifies as important.

Applying Study Skills

1.3 Organizing Textbook Material

Supplies/Prep: Create a concept map or an outline that is partially filled in, one copy per student, and a whiteboard for presentation. Students should bring notes and a textbook.

In STATS 1001, an important study skill is active reading of the textbook. Since the homework questions and most of the material on the exams come from the textbook, having a good understanding of the text is essential. To help the students better organize the information in the textbook, I use two different outlining activities.

Last week, I used a concept map to have them organize one of the central concepts of the chapter. I created a partially filled in map, with the central theme in the middle. The students worked in pairs to fill in the missing parts of the map. I drew a copy of the map on the board, and asked each student to come to the board and fill in part of it, and then explain that part to the class. This week, I had the students fill in a partial outline of the textbook chapter to model one way to take notes out of the book. I filled in some of the outline, and had the students fill in the rest. Both of these activities highlighted effective ways of taking notes, and helped students pull out relevant information from the text.

~ Cameron

Subject Area:

- Any

Grouping:

- Pairs

Notes:

If you repeat this activity later in the semester, prepare just the map and have students generate all of the content, especially main concepts.

1.4 Focusing on Concepts with Note Cards

Supplies/Prep: Blank notecards (4-6 per group). Select specific concepts to have students focus on. They will need their textbook.

In the Sociology PAL session I facilitate, students sometimes struggle recognizing the important themes and concepts of a chapter. When that happens, I like to do an outlining activity that gives them practice in identifying the main ideas. The nice thing about this activity is that it doesn't take a lot of preparation other than knowing which concepts you want the students to identify.

I give each group of students 4-6 blank notecards and ask them to come up with one key concept per notecard from the chapter. Once all the groups have finished, they write their concepts on the board. As a class, we then have a discussion about the concepts, noting similarities and differences between them. Point out repeating themes as they are most likely key concepts. My role is to help students organize smaller ideas under the appropriate main concepts.

This activity can also be used to review before a quiz or an exam. I have done this activity many times as a PAL facilitator and it always goes really well. First year students sometimes have trouble with the sheer amount of reading they have for their classes, but an activity like this helps them to more easily recognize what they are supposed to be learning from a really long chapter as they read it.

~Amber

Subject Area:

- Humanities
- Social Sciences
- Hard Sciences

Grouping:

- Small groups

Notes:

Remind students that a word and definition does not count as an important concept.

Do this activity early in the semester so students get practice identifying main ideas while reading each chapter in their text.

You could take picture(s) of the board work and send them to the students after the session.

Applying Study Skills

1.5 Note-taking in Lecture

Supplies/Prep: Construct a lecture outline with parts left blank (students will fill in), one copy per student. Students will need their own lecture notes or handouts.

Note-taking is one of the first study skills that I model early in the semester. Freshmen often don't take detailed lecture notes because their high school courses never really required it. This activity shows new strategies for note-taking that they may not have seen before and helps them decide if they need to change their note-taking technique. I take notes in a traditional outline form. For this activity I type up my lecture notes in the outline form and selectively delete key words and important definitions. I then hand out the outline and ask students to use their own lecture notes to complete it. At first I have them work individually; after about ten minutes I have them form groups of three to share knowledge and notes to complete any missing parts. The objective is to see how their classmates take notes and hopefully see a new strategy to use in the future. After the groups have filled out the outline, we go over it as a class; usually at least one group has the outline complete.

One time I ran into a slight hiccup that actually worked in my favor. During the first lecture the professor said it wasn't necessary to take notes because the lectures were available online with accompanying slides every week. Even so, I still thought note-taking was an important skill to model. When I introduced the activity in PAL a few days later, the students immediately told me, "But the professor said that it wasn't necessary if we did not want to." I told them that as a student that had taken the course, my lecture notes helped me immensely even though the lectures were posted every week. I also reminded them that they could of course use the lecture slides online, in addition to any notes they had taken, to fill out the outline.

But even with the slides, most students found that they could not fill out the whole outline on their own and some even had trouble completing it in their groups. The activity demonstrated that lecture slides can be vague when trying to recall the whole lecture for studying purposes and that it really helps to take your own notes!

~ Tali

Subject Area:

- Any

Grouping:

- Small groups

Notes:

You might repeat this activity a few weeks later, in case they still seem to resist taking their own notes. Or they might have developed a new strategy that they can share with the class.

1.6 Note-taking in Lecture – 2.0

Supplies/Prep: Generate an outline for students to use during PAL session covering one or multiple lectures.

Over the course of the semester, I found that many students were not paying as much attention to lecture as they should and some were not taking notes at all. To counteract these behaviors, I created a lecture outline. I laid out specific guidelines that they needed to follow and I made the outline fairly vague.

Earlier in the semester we had done a lecture outlining activity that was much more structured and had a lot of “fill in the blank” sections. This time I left the outline much more open so that it required more critical thinking and interpretation. As a result, I noticed more discussion in the small groups – while students completed the outline – and in the large group, when we went over it. In many parts of the outline I asked for examples from the text that reinforced what the professor was saying. Students became even more familiar with the text and they selected relevant quotes, as they would do in their final essay. As a result of doing this in small groups first, we had many more examples to discuss in the large group.

~ Tali

Subject Area:

- Any

Grouping:

- Small groups

Notes:

By using the outline, I was able to expose students to the large concepts discussed in lecture, to which they might not have been paying attention. Also, they now had a model of what good lecture notes looked like, and could use similar outlines for the future.

Applying Study Skills

1.7 Partial Outlines with Textbooks

Supplies/Prep: A worksheet in an outline format for each student; complete one section to use as an example. Use section headings as questions for discussion. Students will need their textbook.

I like to use an outlining activity several times during the semester because it helps simplify the textbook and gives students practice in identifying relevant information. Outlining can be an effective way to take organized notes.

I set up an outline similar to one that I use for my own notes. I use the chapter title as the heading; each of the section headings are Roman numerals I, II, III, etc., the main points are a, b, c, etc.; and the examples become 1, 2, 3, etc. I hand out the skeleton outline with only the first section of the first chapter completed. Students work in pairs, and each pair is assigned a chapter to complete in the outline. Once the outlines are complete, students move to find a new partner who has completed a different chapter. During this portion of the session, the students explain their chapter to their new partner, allowing the partner time to fill out the outline in their own words. We then get together as a large group; we turn each of the section headings into a question. I quiz each group and the students summarize the section when they answer the question. We usually have a good discussion about the chapters and we cover a lot of material.

~Cameron

Subject Area:

- Any

Grouping:

- Pairs

Notes:

Sometimes students try to fill out the outline individually, or just copy their partner's work. To encourage cooperation, hand out one outline per group, and email a completed outline to everybody after the session.

1.8 Venn Diagrams for Concepts

Supplies/Prep: White board and a set of sticky notes with applicable terms.

Study skills such as making notecards and highlighting notes aren't enough to succeed in the upper level Child Psych course that I facilitate. I've learned that the ways students have studied for other classes don't prepare them sufficiently for tests because the professor expects them to think critically and abstractly about the course concepts.

As a result, I design sessions that model new ways to study. For example, in one session, we made a large Venn diagram of three theories of cognitive development to help organize the students' thinking. I wrote a bunch of terms on Post-it® notes and divided them among groups of five students. Students had to decide which Venn circle each Post-it® note belonged and stick it on the board (sometimes it fit in two or all three theories). Once the groups finished, we discussed each term as a large group and rearranged some of the notes. The students spent a lot of time discussing the terms and discussing them more critically and deeply - more than I expected. They were still a little shy and afraid to speak up, so I ended up standing in front of the whole group and guiding the discussion more than I like.

~ Hannah

Subject Area:

- Humanities
- Social Sciences

Grouping:

- Small groups

Notes:

Next time I would allow more time for discussion. I would tell students earlier in the session that they will come up to the board to explain the terms and rationalize where they placed them.

Applying Study Skills

1.9 Organizing with Timelines

Supplies/Prep: Whiteboard required for presentations, students should bring textbook and notes to the session.

For the topic of Piaget's theory of cognitive development, I created a timeline activity to introduce students to a visual way of organizing information. I drew a very long line on the board and divided it into four sections, with each corresponding to one of Piaget's four stages.

I asked students to form four groups and assigned each group a stage. Their task was to write on the board key words and characteristics of their assigned stage and present the information to the class. Presenting the information got everyone to participate and gave students time to copy the timeline in their notes.

When I had made my own timeline in preparation for the session, I was surprised how helpful it was for me to remember and understand the different stages. I hope that's a sign that my students found it helpful as well. After the activity, I had them talk in their groups to apply what they had learned about Piaget's stages to Kohlberg's stages of moral development. I overheard some good questions and discussions as they tried to process and blend these different theories – a good sign that things were "clicking".

~ Mindy

Subject Area:

- Humanities
- Social Sciences

Grouping:

- Small groups

Notes:

Give students time to take notes on the material that the rest of their classmates presented.

1.10 Peer Reviews of Essays

Supplies/Prep: Copies of “Guidelines to Writing Good Thesis” handout for each student (see back).

In the journalism course I facilitated, I chose an activity related specifically to writing an academic paper. The session focused solely on writing development and peer review. I started with the "Guidelines to Writing a Good Thesis" handout, which covers the purpose of a thesis, how to write one, and where to place it in the paper. Each student was given a peer review worksheet that focused on seven aspects of writing an academic paper: organization; development and support; style; grammar and mechanics; introduction; body; and conclusion. Working in pairs, students exchanged papers and answered a series of questions regarding their peer's paper. The peer reviewers had to reflect on the actual content of the paper to see if the writer stayed focused on the topic and supported the introduction with the ideas in the body. By way of the worksheet, the peer reviewer gave feedback on the strengths and weaknesses of the paper.

~ Samantha

Subject Area:

- Humanities
- Social Sciences

Grouping:

- Pairs

Notes:

Students griped about having to answer each of the questions and use support and examples from the paper.

In the end, they appreciated the activity because they had many thoughtful comments on their papers that focused on important aspects of good writing.

Applying Study Skills

➤ **Handout: Writing a Good Thesis** *(developed by facilitators in U of MN PAL program)*

A thesis should fit the following criteria:

- Tells the reader how you will interpret the significance of the subject matter under discussion.
- Is a road map for the paper; tells the reader what to expect from the rest of the paper.
- Directly answers the question being asked. A thesis is an interpretation of a question or subject, not the subject itself. The subject, or topic, of an essay might be World War II or Moby Dick; a thesis must then offer a way to understand the war or the novel.
- Makes a claim others might dispute.
- Is often a single sentence somewhere in your first paragraph that presents your argument. The rest of the paper (the body) gathers and organizes evidence that will persuade the reader of the logic of your interpretation.

Do I have a strong thesis?

- *Do I answer the question?*
Re-reading the question prompt after constructing a working thesis can help you fix an argument that misses the focus of the question.
- *Do I take a position that others can challenge or oppose?*
If your thesis simply states facts that no one would, or even could, disagree with, it is possible that you are providing only a summary, rather than making an argument.
- *Is my thesis statement specific enough?*
Vague thesis statements lack a strong argument. If your thesis contains words like "good" or "successful," see if you could be more specific: why is something "good"; what specifically makes something "successful"?
- *Does my thesis pass the "so what?" test?*
If a reader's first response is, "So what?" then you need to clarify, forge a relationship, or connect to a larger issue.
- *Do my arguments support my thesis specifically and without wandering?*
If your thesis and the body of the paper do not go together, one of them has to change. You may change your working thesis to reflect new ideas you developed while writing the paper. Reassess and revise your writing as necessary.
- *Does my thesis pass the "how and why?" test?*
If a reader's first response is "how?" or "why?" your thesis may be too open-ended and lacks guidance for the reader. What can you add to give the reader a clearer sense of your position right from the beginning?

1.11 Post-Exam Reflections

Supplies/Prep: Formulate a post exam reflection worksheet. Print one per student.

One activity that I like to do is have students complete a post-exam reflection after each test. Students in Statistics 1001 have five exams during the semester and I have used this reflection after each exam. The reflection asks students how they studied, what specific skills they used to study, and how much time they spent preparing for the exam. It also asks them to predict their score on the exam. I ask them to put their names on the reflection and I collect them. I return them before their next exam so they can see how their expectations matched with their actual performance. Then they can review what study skills they used in order to evaluate what worked and what didn't. Looking through the reflections throughout the semester, I have seen some students change both how they study and how much time they spend preparing for the test.

~ Cameron

Subject Area:

- Any

Grouping:

- Individual

Notes:

It has also been really beneficial for me to see their progress and also to see that PAL is helping them improve their study skills and find what works best for them.

This activity doesn't take the entire class time and should be paired with another activity.

A similar post exam activity is found in 9.3.

Applying Study Skills

➤ **Handout: Post-Exam Reflection** *(developed by facilitators in U of MN PAL program)*

1. The information that you remembered best on the exam was learned by:

2. How confident were you walking into the exam? Why?

3. Which of the following activities did you do **before** the exam? (Check your responses)

- ☐ Studied with other students in the class (outside of PAL)
- ☐ Attended more than one PAL session
- ☐ Studied the online exam study guide
- ☐ Read all assigned readings at least once
- ☐ Read assigned readings prior to their associated lectures
- ☐ Reread/skimmed readings before the exam
- ☐ Took written notes about the assigned readings
- ☐ Highlighted assigned readings
- ☐ Made comments in the margins of assigned readings
- ☐ Printed or downloaded the lecture PowerPoint slides ahead of class
- ☐ Took notes during class or added additional comments to the slides
- ☐ When taking notes, rephrased what the professor said in my own words
- ☐ Reread my lecture notes again before the exam
- ☐ Created outlines for each of the potential essay questions on the study guide
- ☐ Defined each of the vocabulary words listed on the study guide
- ☐ Predicted potential exam/multiple-choice questions based on my lecture
- ☐ Practiced writing answers to my potential essay questions

4. About how many hours did you spend studying specifically for this exam? ____

5. How many class sessions did you miss *(for any reason)* before the exam? ____

6. Which of the following activities did you do **during** the exam? (Check your responses)

- ☐ Made an outline of essay responses before writing them.
- ☐ Marked up the exam questions by underlining or circling key words and phrases.
- ☐ Crossed out incorrect answers in multiple choice items.
- ☐ While answering essay questions, looked back at the multiple-choice questions to find more information to incorporate into my answers.
- ☐ Watched the clock to make sure I had enough time to complete the entire exam.

7. How did you feel about your performance when you left the exam?

8. How do you think you did on the exam? (Can give a %, letter grade, or general idea)

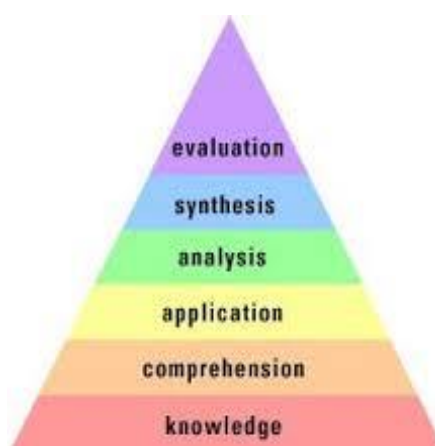
SECTION 2: Grasping Concepts

The activities in this chapter not only help students learn the material, they also challenge students to take their knowledge a step further and apply it to an unfamiliar scenario. They target the “application” and “analysis” levels of higher order thinking represented by Bloom’s Taxonomy.

Worksheets often require memorization and recall, along with a basic understanding of facts and ideas – thinking skills that rank lower in Bloom’s pyramid. While this is beneficial, in order for students to truly have a deep understanding of the material, they need to be able to apply their knowledge to new situations – ones that are not so straightforward.

This set of activities is best used when students have studied and feel comfortable with the material and are ready to think outside the box to solve problems or make connections between seemingly different sections of the course.

~Justin



The six levels of
the cognitive domain
of Bloom’s original taxonomy

Grasping Concepts

2.1 Defining Terms

Supplies/Prep: Printout of 5 column chart (1 per pair of students); students should bring textbook and lecture notes.

I often do an activity in my Child Psych PAL sessions that helps students integrate textbook material with the lecture. It's especially useful because the professor expects students to know material from the textbook but does not go over it explicitly during lecture.

I create a chart with five columns labeled: Term; Textbook Definition; Lecture Definition; Combined Definition; and Example. I group students into pairs and instruct one person in each pair to work with the textbook and the other to use their lecture notes to complete the chart. After 10-15 minutes, I ask the pairs to re-convene and compare the terms they found. If the definitions vary, they are to come up with a definition for the term that combines both the text and the lecture version into one coherent definition. The examples in the last column must still relate to their final definition of the term.

Students tell me they really like this activity because it helps them organize the information in their brains. Sometimes it is hard to understand a term when two sources used in the course have similar but not exactly the same definitions. The activity requires students to look at concepts from two different points of view and then asks them to explore a third point of view in the form of their own wording/ideas. It definitely stimulates a deeper understanding of course material.

~ Hannah

Subject Area:

- Social Sciences
- Hard Sciences

Grouping:

- Pairs

Notes:

Sometimes the professor takes the definitions straight out of the book; in that case, this activity wouldn't be as useful. Still, students can think about examples that describe the term.

2.2 Grasping with Graphs

Supplies/Prep: print 5 different graphs - one set per group; tape; two data sets for groups to analyze.

I facilitate a statistics course. For the final session I planned a graphing activity. I usually design a worksheet for the students but this time, I wanted things to be a bit more interactive. I split the students into four groups and gave each group the same five graphs to analyze. These graphs were taped up on the wall in different areas of the room. I asked the students to determine the type of graph, what data was being displayed, and what issues, if any, there were with the way the data was presented. In order for me to quickly assess whether there was confusion or not, I had each group write their answers on the whiteboard. As a large group, we discussed any common problems the students had with the analysis.

For the next part of the activity, I gave the groups two data sets and let each group decide how they would present the data. This created some good discussion within groups because someone would suggest one type of graph, and another member would counter that idea with their own recommendation. Seeing this discussion play out let me know that they were critically thinking about the problem.

~ Andrew

Subject Area:

- Hard Sciences

Grouping:

- Small groups

Notes:

Remind the students not to be persuaded by other groups' answers on the board and to remain confident in their answers. Varying answers provides for more opportunities for discussion.

Grasping Concepts

2.3 Making Connections

Supplies/Prep: Each group will need access to a computer, textbook, and class notes. Prepare questions that reflect lecture topics.

As a facilitator for a political science class, I found it's really important to connect concepts and facts from the course to the real world. It was easier to do when the students were studying the presidency, the courts, and Congress, but was harder when studying public opinion and the political parties. But since it was important, I tried to include real-life examples in my PAL sessions. I did this when the class was learning about government bureaucracy and special interest groups.

I had students form groups of 2-3 and, using a computer, books, or their notes, asked them to pick a special interest group. For the first few minutes they researched the history and goals of that interest group. Once they had established the background, I posed some general questions to connect their examples with the concepts being learned. After each group completed their research, students presented their findings to the whole class. This proved to be an excellent activity to help solidify concepts and facts from the course and see them at work in real-world examples.

~ Trent

Subject Area:

- Humanities
- Social Sciences

Grouping:

- Small groups

Notes:

I kept a close eye on students while they used their computers in case they got distracted and off topic.

2.4 Using Outside Readings

Supplies Prep: Identify literature that would be applicable to current topics covered in the lecture. Generate a set of questions for students to discuss after reading.

For one of my anthropology PAL sessions, I had the students read an ethnography about Americans that was written as a satire. I chose this piece not only because it was funny, but because the students were already very familiar with American culture and would be able to analyze the author's representation in a way that they could not with some of the course texts. I wanted the students to use this text to recognize some of the problems ethnographies present, as well as articulate the benefits and deficiencies of ethnographic writing. After the students had a chance to read the text, I posed some general questions regarding how ethnographers represent their subjects and whether or not their writings lead to a better understanding of a culture.

~ Alina

Subject Area:

- Humanities
- Social Sciences

Grouping:

- Small Groups

Notes:

Choosing a good text is important – it needs to address the themes and concepts discussed in lecture. The questions also need to emphasize this connection.

Structure the discussion so quieter students will participate by assigning specific questions to each group.

Grasping Concepts

2.5 Learning in 3-D

Supplies/Prep: Yarn and worksheets with 10 multiple choice questions (optional).

I facilitate an Introductory Biology PAL session and have found that a lot of the molecular biology material can be difficult for students to grasp. However it helps if I can replicate cellular concepts using a larger scale. This semester I did an activity similar to one my high school Biology teacher did for the concepts of mitosis and meiosis. First, I used yarn to make “cells” and “chromosomes.” Then I put the students into groups of two and had them work with each other to learn the steps of mitosis and meiosis using the yarn as a model. I employed Socratic questioning to help students think critically about what they were doing. They seemed to really enjoy working through the steps together in this modality of learning.

~ Tara

Subject Area:

- Hard Sciences

Grouping:

- Pairs

Notes:

This activity didn't take long, so I also put together a set of ten multiple-choice questions that I had created about mitosis and meiosis.

If the yarn isn't needed, let students take the yarn models home if they would like to continue practicing.

2.6 Making Space for Media

Supplies/Prep: Online resources such as YouTube or related video and/or webpage (optional).

Postmodernism is a complicated theme and even students ready to graduate can have a difficult time understanding this topic. Because of the complexity, I wanted to design an activity that would allow students to create their own definition based on their prior knowledge, as well as mix in a little comic relief.

I began by having the students define modernism from what they had learned earlier in the semester. With this background set, I facilitated a discussion to define post modernism by comparing and contrasting the two ideas on the board. I chose this approach because I was aware of an online resource that had the same comparison and had good definitions for both terms, which I showed at the end of the session. As the students began to understand more about postmodernism, I had each group create their own definition together.

For a humorous slant on this challenging topic, I showed a YouTube clip of a comedian commenting on the phenomenon of post modernistic thought. After hearing this more abstract idea, I wanted the groups to try to redefine their idea of postmodernism as well as think of five different ways in which daily life changed from the modern era to the postmodern era.

~Mercedes

Subject Area:

- Humanities
- Social Sciences

Grouping:

- Large group

Notes:

Because the topic was complex, this activity worked well as one large group. Students could both build on, as well as challenge, each other's ideas.

SECTION 3: Promoting Participation

Group work is the foundation of peer-assisted learning. In our PAL program “peer” refers to the students sitting next to each other at the session. Instead of having the facilitator stand at the front of the room answering questions – as the knowledge expert – students learn from each other through collaboration. The key to making this happen is to develop good group dynamics.

But what if the students attending your session are shy and don’t know each other? What if they are habitually unprepared and don’t participate for fear of exposing their lack of attention. Maybe they are not used to or don’t like this kind of learning. When group members won’t talk and share their thinking with each other, not only does the activity lose its effectiveness, the students’ learning suffers. We find these challenges are more likely to occur in the PAL sessions where attendance is required.

The activities in this chapter are focused on getting students to interact more readily; sometimes it’s as simple as using an ice-breaker or designing an activity where everyone has to speak. It might be a matter of using a different grouping strategy. Many of these activities are best suited for the beginning of the semester or if the group dynamic seems to fall in a rut midway in the term.

~Justin

More than anything, groups need clear goals, appropriate material, adequate support, and solid leadership. We have seen these four factors... make the difference between a group whose members feel enthusiastic, contribute actively, and learn at deep levels and a group whose members see the group work as yet another task to complete, unsure of exactly why they are participating.

– Gregory Light, *Making Scientists: Six principles for effective college teaching*

Promoting Participation

3.1 Quiz Quiz Trade

Supplies/Prep: Generate a set of questions that pertain to the material – make sure there are at least as many questions as there are students. Put these questions on small strips of paper or notecards.

For my Math 1051 PAL sessions, I like to use *Quiz Quiz Trade* as a warm-up. I prepare strips of multi-colored paper – each piece of paper has a math problem to solve or a conceptual question to discuss. Each student receives a piece of paper. I instruct them to find a partner, and take turns asking the question on their slips of paper to their partners. Afterwards, they swap questions. Then, each person, with a new slip of paper, finds a new partner. The process repeats with partner pairs quizzing each other and trading papers.

Quiz Quiz Trade can take up very little time, or it can span the time of an entire activity – it is very versatile! It is up to you to decide how long students move around and interact with each other. However, at the end, debrief the questions as a large group. You might ask students “What are the highlights from your discussions in this activity?” Allowing time for large group discussion is important because it gives students the chance to unravel their thoughts with their peers.

Encourage students to get out of their seats and move around. The point of this activity is to promote participation and allow students the opportunity to work with new people.

~ Jane

Subject Area:

- Any

Grouping:

- Pairs

Notes:

You might want to come prepared with extra questions for this activity. If a student receives a question for a second time, they can swap it out for another.

I find it helpful to be very explicit with this activity – I tell students “This activity is designed to get you all up and moving.”

3.2 Rotating Worksheets

Supplies/Prep: Printed worksheet for each student

The most common problem I had during my PAL session was students not actively participating in the group activity. If the activity was a worksheet, some students would not even bother to fill it out; others would race ahead of their group members to complete the worksheet just for themselves and not help other members who were stuck on a question. To remedy this problem, I created a new format to promote more group interaction and more accountability to the group.

I put students into groups of four and gave each student a worksheet. As a group, the students discussed the first question and came to consensus on a good answer, which they all wrote down in their own words. After completing the first question, each group member passed their worksheet to someone else in their group, so no worksheet was completed by just one student. At the end of the session, each student left with one of the completed worksheets.

My goal in structuring the activity this way was to promote active listening while discussing answers since each student had to write the answer in their own words. If students were not actively engaged in the discussion, the quality of their answers on the rotating worksheets would be poor for everyone. The other goal was to keep each group member working on the same question (to keep any individual from rushing ahead) since the worksheet wouldn't rotate to the next person until everyone had finished writing.

Overall, this activity worked really well. I heard or saw everyone participate (or at least listen) and group members worked at the same pace and helped each other along. Another unforeseen benefit of the rotating worksheets was that the students became more motivated to do quality work since their peers would be reading what they wrote.

~ Tali

Subject Area:

- Humanities
- Social Sciences

Grouping:

- Small groups

Notes:

An adaptation of this activity is to give each group only one worksheet and have each member take turns recording the answer.

Promoting Participation

3.3 Unprepared!

Supplies/Prep: None (this activity is a handy backup if students come unprepared).

This past week, I tried to correct a problem I have experienced in my PAL sessions for the past two weeks. In the English literature course I facilitate, students are assigned several chapters of reading each week; I usually follow by creating questions related to the weekly reading. However, most of the students in my session have stopped doing the reading, and are unable to discuss the questions. We struggled through last week's session as they aimlessly flipped through their book trying to find the answers.

This week, I preemptively planned an activity that would be beneficial even if they had not done the reading. At the beginning of the session, I assigned each student two chapters. I asked them to skim the chapter, highlight the main plot points, identify themes presented in the section, and record related details. Each student then presented their section by writing it on the board; the rest of the students were instructed to make notes on everyone's presentations.

~ Jennifer

Subject Area:

- Any

Grouping:

- Individual

Notes:

At the expense of peers working together as a group, I reasoned that we could cover more material if they worked alone. If you have a big session, working individually might not be useful.

I still remind the students that they will need to read the book for the exam.

3.4 Warming Up for Discussion

Supplies/Prep: A set of questions based on the writing assignment; extra questions about the reading (in case discussion dies out too soon).

The English literature course I facilitate is focused mainly on discussion and writing assignments. A welcoming PAL session is a perfect place for hesitant students to begin to express their thoughts in front of their peers without judgment from someone who grades them. It's also a good opportunity to bounce writing ideas off one another. With that in mind, at the beginning of the semester, before their first writing assignment, I give students an opportunity to practice/rehearse expressing their ideas for the TA discussion section and to better understand how to approach their writing assignment.

To accomplish this, we arrange the tables and chairs to have one large group discussion. I prepare questions from the readings that would be similar to what a TA might ask and then I let the students have the floor. I also make sure that each person contributes at least once, and if they aren't talking, I look for signs of active listening (i.e. eye contact with the speaker, nodding, and small facial expressions).

To prepare for the writing assignment, I break down the writing topic into smaller questions for everyone to discuss. I know this format is successful because I often hear them articulate thoughtful analyses that go beyond what the question is asking. It's great to see students take notes while they are listening.

~Amanda

Subject Area:

- Humanities
- Social Sciences

Grouping:

- Large group

Notes:

Getting students to contribute to discussion can be difficult sometimes. I've learned to not be afraid of the long silent periods. Often, students are planning what and how to voice their ideas, so it's important not to rush.

Promoting Participation

3.5 Flexible Grouping

Supplies/Prep: Printed worksheet for each student.

Because I lead a voluntary session for an upper level course, the attendance varies from week to week; sometimes I have one student, other times 15 will show up. Plus, upon the professor's request, I hold large group review sessions before exams; one time I had 56 students attend. Because I have to plan my activities to accommodate a broad number of students, I use flexible grouping. For example, I often create charts (on worksheets as well as on the whiteboard) to demonstrate different stages in a psychological theory. If 12 students come, then I divide them into three groups of four. Each group does just one part of the activity, and then I have them share their portion with the rest of the groups. If only four students come, I usually ask them what they prefer to do – divide up or work as one group. Usually they want to work as one group, so we walk through the activity all together.

~ Hannah

Subject Area:

- Humanities
- Social Sciences

Grouping:

- Varies

Notes:

Neither grouping strategy is really any better than the other; if it's a large group, they have a chance to explain concepts to each other, and if it's a smaller session, they get to hear from everybody in the group, as well as answer my probing questions.

3.6 Student-Led Discussion

Supplies/Prep: Create 10 or more questions about the material being reviewed. Prepare a PowerPoint presentation, putting one question on each slide (no answers); bring laptop; classroom equipped with a projector; candy – optional).

To encourage students to take more ownership of the session, I tried a new activity in my Psychology PAL session. I brought my laptop to the session and explained to students that they would take turns coming up to the computer to type in answers to questions that were based on the course study guide. The person typing was really just a recorder; the answers would be supplied by their classmates. I also told them that I would email everyone the finished PowerPoint so they could more easily participate in the discussion instead of feverishly writing down what was being typed. To further entice students to participate, I brought candy. Since each student only typed the answer to one question, most of them got a turn at the computer. Those who didn't type were fairly involved in the discussion, so the activity allowed for a wide range of participation. I was impressed with how well it promoted really good discussion; plus, students generated real-world applications of the concepts.

I emphasized that the typists didn't have to know the exact answer; they only had to summarize the information their peers gave them. I think this helped the students feel less pressure about being at the computer, in front of the class. There was one student, however, who did not put in much effort. It might work to make the activity into a game if there are more students like that.

~Jeanna

Subject Area:

- Humanities
- Social Sciences

Grouping:

- Varies

Notes:

I also told the students right at the beginning that I would remain mostly silent during the activity; it was their responsibility to generate ideas. However, if they did start to discuss something that was incorrect, I would stop them and give them hints to get back on track.

Promoting Participation

3.7 Flexible Topics

Supplies/Prep: Organize problem sets for at least four separate topics or concepts. Prepare worksheets of problems for each topic. Each table is a different “station”. Place two copies of each set of problems at a station. Give each student two index cards.

At the last PAL session before the math midterm, I decided to more actively engage the students; I wanted to do something more than just my usual simple worksheets. My sessions were located in a large classroom with long tables and students were stuck in a routine of working with just the people next to them. I designed an activity that would get the students out of their seats and moving around.

First, I had students write on an index card what they thought was important to study for that midterm; they could use their textbooks and lecture notes. The index card and their classmates would be the only resources available to them during the PAL session. Knowing the importance of self-monitoring one’s learning, I asked students to prioritize the topics they wanted to practice; it was up to them to determine their weak areas and to choose stations related to the topics they felt were most challenging.

At the end of the session, I gave each student another index card and asked them to write down any new information that was missing on their original card so that they could refer to both index cards when studying for the midterm. Students liked having more flexibility and control of what to work on in the PAL session, instead of doing whatever problems I gave them.

~ Kaiwen

Subject Area:

- Math
- Hard Sciences

Grouping:

- Individual

Notes:

If I did this activity again, I would tell students ahead of time that they should not spend more than 10 minutes on a single problem since it would not be a wise way to manage their time on the exam.

This activity works best with a larger class size – students are more likely to find classmates working on the same set of problems. My rule of thumb is the total # of students should be > the # of stations.

SECTION 4: Solidifying Material

This chapter focuses on activities that facilitate the learning of brand new material, as well as activities for that sweet spot after introduction of the material, but before exam review. There are times when you can't quite decide if the students are comfortable with the material or not. Maybe some are caught up on all the lectures, but others are behind, and there is no pressing exam coming up in the course. These activities are designed for a wide range of participation to help cement ideas into the students' repertoire of knowledge. For example, say the instructor has fallen behind in the syllabus; the options for the week's PAL session are to either review the material that you worked on last week, or move ahead to what the professor just lectured on yesterday.

In my experience as a facilitator, I have found that students are either unable, or simply choose not to review and fully comprehend the previous day's lecture. If you decide to move forward with your PAL sessions anyway, an activity based on the new material will probably be a huge bust. While there is no right or wrong decision (having another review or moving on to new material), this chapter offers ideas on how you can help guide students through the most current lecture material when they don't understand it.

~Justin

People who learn to extract the key ideas from new material and organize them into a mental model and connect that model to prior knowledge show an advantage in learning complex mastery.

– Peter Brown, *Make it Stick: The science of successful learning*

Solidifying Material

4.1 Go with the Flow

Supplies/Prep: Prepare 8-10 text analysis questions; important quotes to discuss from the text; some questions to stimulate large group discussion or debate (optional).

I facilitated a PAL session for a U.S. History course on the day after spring break; the students were to have finished the book *Negroes with Guns* by Robert Williams and compose an analysis of the book. To guide them on how to approach their analysis, I planned two separate activities.

In the first activity, I gave students eight questions to discuss in small groups. The goal was to have them consider ways in which the book provided historical context for what they were learning in class. While they were discussing in small groups, I wrote a quote from the book on the board for them to consider: “The Negro in the South cannot expect justice in the courts. He must convict his attackers on the spot. He must meet violence with violence, lynching with lynching.”

After each of the groups finished talking about the questions and the quote, I opened the discussion up to the entire class. This was the second activity and initially, it was supposed to be a mini debate based on the quote and the question, “Do people have the right to defend themselves no matter their race?” The goal was for the students to identify which questions they should consider when writing their analysis paper. Instead of having a debate, the students brought up the recent Trayvon Martin case and connected it to self-defense and the author Robert Williams. Though we strayed from my original agenda, the session was still successful in achieving the intended goal; it allowed the students to make historical and current day connections that would be helpful for their analysis.

Subject Area:

- Humanities
- Social Sciences

Grouping:

- Small groups

Notes:

As a facilitator, it is important to design a flexible agenda; if an applicable discussion arises like in this session, don't be afraid to run with it.

~Mercedes

4.2 Exploring Themes

Supplies/Prep: Prepare examples that connect the themes using various graphic organizers.

For one of my English Literature PAL sessions, my goal was to have students understand 1) how themes work together to make a single story cohesive and 2) how different stories can be connected through similar themes. To do this, I instructed the students to write down themes they had encountered so far in the course. While they shared them, I wrote the themes on the board.

I asked each group (of 2-3 students) to pick one theme from the board and connect it to the other themes using some kind of graphic representation (flow-chart, pictures, Venn diagram, etc.). Next, each group shared their visual representation and explained how they had arrived at their scheme. Surprisingly, each one was different. A lively and thoughtful discussion ensued when each group tried to justify their arrangement of themes. To help cement these ideas, I asked students to write possible essay questions/topics centered on the themes. Everyone was able to generate at least one solid question by the end of the period, showing me that they had benefited from the exercise and the discussion.

~Amanda

Subject Area:

- Humanities
- Social Sciences

Grouping:

- Small groups

Notes:

Whether you share the examples you prepared ahead of time probably depends on how the activity goes.

Solidifying Material

4.3 Brainstorming Themes

Supplies/Prep: Determine three themes from lectures and readings for students to focus on.

The plan for one of my American History PAL sessions was to have students think about three major themes that have shaped how we view our country's history. I created this activity because their course final was an in-class essay; I knew from having taken the course that it would be useful to have a number of events to pull from that would support their arguments in the essays.

For each theme, students were to choose three events in American history that shaped and defined the theme, i.e. freedom. Students worked in pairs to discuss their different viewpoints about each event and how it related to the theme of freedom.

Midway through the session I realized that many students did not have a good grasp on how events shaped history. My original goal was one they were not ready for. Unfortunately, I had assumed students better understood how events influence the way we view history. They suggested that we work from a list of events and match them with the themes and discuss them as a group; I agreed, but I made them compile the list.

~ Mercedes

Subject Area:

- Humanities
- Social Sciences

Grouping:

- Pairs

Notes:

This experience showed how important it is to be flexible during the actual session, and to be ready with back up plans if the activity isn't going well.

The facilitator's knowledge of the material allowed her to quickly come up with examples to get students started.

Doing the activity ahead of time, as if the facilitator is the student, is never a waste of time!

4.4 Journey to Physiology Paradise

Supplies/Prep: Determine learning objectives (LO) applicable for the week's material; create test questions for each LO. Make a grid (see picture) on laptop; room with projector.

This activity works well in courses that focus on specific learning objectives or tasks. For example, in my physiology course, every lecture has 5-10 corresponding LO's. Every test question comes from a specific LO. This was a great way to model how studying the LO's will help them prepare for exams.

The goal of the activity is to navigate the grid from the start box to "Physiology Paradise" (or whatever corny name is applicable to your course). Each square on the grid is assigned a lecture (or topic). Students move around the board by studying a given learning objective pertaining to that lecture and correctly answering sample test questions generated from that learning objective.

Working in groups of 3-4, students picked out a color to mark their starting point on the grid. I gave them a LO (from the lecture associated with the spot on the grid), which they discussed in their group until they felt they had "mastered" the material. I then asked them a question from a sample exam that was generated from the LO they were studying (see sample). If they answered the question correctly, they "won" that spot and I colored it with their team color. If not, they were given a new LO and tried again. Teams could move in any direction but not on squares already won by another team.

This activity could be altered for other courses as well. For a math class, I might write the learning objective as, "Simplify a rational expression with variables in the exponents." Students would discuss the applicable rules or properties and then I would give them a problem to solve.

~Elaina

Subject Area:

- Any

Grouping:

- Small groups

Notes:

Most of the prep time was spent matching questions to learning objectives. It would help to have the questions typed up separately and coded to the appropriate learning objective.

Solidifying Material

START							
MN	S	AP	NS	MN	S	AP	NS
NS	MN	S	AP	NS	MN	S	AP
AP	NS	MN	S	AP	NS	MN	S
S	AP	NS	MN	S	AP	NS	MN
MN	S	AP	NS	MN	S	AP	NS
NS	MN	S	AP	NS	MN	S	AP
AP	NS	MN	S	AP	NS	MN	S
S	AP	NS	MN	S	AP	NS	MN
Physiology Paradise							

MN = MetaNeuron

S = Synapses

AP = Action Potentials

NS = Nervous System

(These all corresponded to their lectures and lab from one week.)

If a group picked an "AP" spot, I might give this learning objective:

"Define **relative** and **absolute refractory period** and explain the state of ion channels during each."

This is a sample question I might ask:

During the relative refractory period, a larger stimulus is required to generate a second action potential because _____ at this time.

1. Most voltage-gated Na⁺ channels are still inactivated.
2. K⁺ leak channels are closed.
3. Voltage-gated K⁺ channels are still open.
4. Voltage-gated Na⁺ channels are still open.

(Learning Objective and Sample Test Question by Dr. Kierstad)

4.5 Engaging with Real Life Data

Supplies/Prep: Warm-up problem(s) written and printed on a worksheet (1 per student).

One of my first PAL sessions for Statistics was focused on learning how to create stem & leaf plots, as well as histograms. There were several problems in their homework that addressed these types of graphs, so I wanted the students to feel comfortable making them and to understand the appropriate situations in which to use them. I chose a sample problem from the book and broke it into four steps: in the first two steps students made a couple of stem and leaf plots with different length stems; the third step was creating a histogram; and the fourth step was simply reflecting on the best way to display the data.

Next, we generated a real-life data set in the PAL session, based on the number of siblings each student had. With the data, the students made a stem and leaf plot as well as a histogram. I thought that the introductory four-step problem was going to be too dull and simple; however I learned quickly that the students did not understand the material as well as I thought. It turned out to be a great way to start because, during the sibling data work, students analyzed the data more quickly and with a better understanding of the various graphical representations. They really enjoyed generating their own data set – it was more engaging and kept them from getting too bored just doing worksheet problems.

~Andrew

Subject Area:

- Social Sciences

Grouping:

- Small groups

Notes:

For a basic concept like this, lots of repeated practice and exposure with different data sets seemed to be the most helpful.

SECTION 5: Sharpening Problem-Solving Skills

The activities found in this chapter are more adaptable to PAL sessions supporting math and science courses. Problem-solving skills in these courses are the key to doing well on exams and homework. Just a few of the common difficulties students often encounter are: (1) not knowing where to begin or even what equation/formula to use, (2) knowing where to start but getting stuck midway through the process, and (3) not having a sense of what the solution should look like. Certainly more can be added to this list.

To practice problem-solving skills, students in these activities are actively involved in showing and explaining their work, step-by-step, either on the worksheet or on the whiteboard. Facilitators have even incorporated technology to aid the process. In some cases, the room set-up offers students a choice to work on the types of problems that are giving them the most trouble. The main goal of these activities is to provide an opportunity to experience the struggle and share problem-solving strategies. The PAL session is meant to be a safe place to make mistakes, which often provide the best learning.

~Justin

A good discussion of problems can foster critical and creative questioning skills among students, skills that students can and do internalize in their own thinking. Problems that do not promote discussion... are unlikely to promote that critical, metacognitive level of internal discussion at the heart of thinking in the scientific mind.

– Gregory Light, *Making Scientists*

Sharpening Problem-Solving Skills

5.1 Clockwork Math

Supplies/Prep: Prepare 12 problems, each on a separate sheet; tape each sheet to the walls around the room.

I like to use "Clockwork Math", which is similar to *Station to Station* (6.5). I split students into 6 groups and they move around the room clockwise doing problems.

I start groups at problems 1, 3, 5, 7, 9, and 11. The groups complete three problems (i.e. group 1 solves problems #1, 2, and 3), putting the solutions on the board. When all groups have solved their three problems, everyone stops. Each group presents the problem they just finished to the whole session (group 1 presents #3, group 2 presents #5, etc.). They also have to make sense of the solutions left by the previous groups (group 1 will also have the solution that group 2 did, since group 2 started on problem #3), figure out which is correct if they are different, and then present it. Continue around the clock with the same pattern.

Getting students up and out of their chairs can make a HUGE difference in participation! I notice they talk a lot more and feel less self-conscious; everyone is a bit louder but only the people in their group are really listening.

~ Lauren

Subject Area:

- Math
- Hard Sciences

Grouping:

- 6 groups total

Notes:

To shorten the activity, you could repeat problems #1-6 in place of problems #7-12, making only 6 problems total.

Always have solutions worked out on paper before the session.

Use problems that generally take an equal amount of time to complete.

5.2 Challenge by Choice

Supplies/Prep: Select three problems for each topic/concept being reviewed. Organize problems by topic/concept; print two copies of each sheet (of three problems) for each topic; title sheets for each topic/concept using a large font; tape.

One activity that did not go well for me was a "stations" activity. Each station was a different topic that students had been learning in an introductory chemistry course. Working in pairs, students were to choose the toughest topics – ones that they struggled with the most – and work on the problems at those stations.

Using the walls in the room and in the hallway, I taped the sheets around the room so each wall was a station for that concept. I made a second copy of each sheet in case there is a crowd at the station.

The first mistake was the size font I had used on the worksheets was too small! Another thing I noticed was that students seemed less interested than usual, when they were just working at their desks. Maybe being up and walking around had something to do with it. They actually kept moving around at a good pace, which was something I had worried about. For the last ten minutes, we went over the answers to the questions that had given them the most trouble.

~ Justin

Subject Area:

- Math
- Hard Sciences

Grouping:

- Pairs

Notes:

I recommend that the facilitator plans how students will move between stations, such as by time or song instead of just randomly. On the other hand, I would hesitate to force students to move on to the next station before they've finished a problem; we're trying to build persistence as well.

For variations of, and different experiences with, "stations"- type activities see stories:
3.7 Flexible Topics
5.1 Clockwork Math
6.5 Station to Station

Sharpening Problem-Solving Skills

5.3 PowerPoint: Is it worth it?

Supplies/Prep: Create a PowerPoint slide show of problems, with one problem per slide. Format it so that the first slide shows the problem and the next slide reveals the solution.

It was my third session as a Chemistry PAL facilitator and I was finally starting to feel more confident. I prepared a PowerPoint presentation with problems for students to work on in groups. Each slide contained a single problem. After each problem, one student from every group wrote their answer on the board and the steps they completed to reach it. It was good for students to see how other groups had solved it, and sometimes they would catch mistakes in their own work after seeing another group's work. However, this often backfired on the students; groups would change their work to match another group's work, even though the other was incorrect and their original solution had been right.

For a group of this size (20 students), I recommend making a simple worksheet instead of a PowerPoint. I like to float around the room and sometimes sit in the back, but it was difficult to do that since I was up at the laptop controlling the PowerPoint. However, PowerPoint is great for displaying the steps in solving long problems because it is much faster than having to write it all out and more advantageous when group sizes are larger and there is insufficient time (or board space) to have students write their work out on the board.

~ Samara

Subject Area:

- Math
- Hard Sciences

Grouping:

- Small groups

Notes:

Next time I will stress that student leave their answer as is and not copy their peers' work. It's just as important to discuss wrong answers as correct ones, since both provide insight into understanding the problem.

The facilitator should remind students that PAL is a safe, welcoming environment and it is okay to not know or have the wrong answer.

5.4 Tutor-Teacher Pairs

Supplies/Prep: Create a worksheet for each student. Format it so paper can be cut in half with an equal number of problems in each half. Cut and bring both halves to class.

As a math facilitator, I tend to fall into a pattern of doing the same old worksheet routine every week. But sometimes a change of pace is needed to boost participation. My favorite activity for changing up the routine is to have pairs of students assume tutor/teacher roles.

I split the students into two equal groups and hand out the top half to one group and the bottom half to the other group. For the first part of the session, students with the same half sheet work on the problems in small groups. When everyone is finished, I ask them to find a partner who has completed the other half of the worksheet. Instead of reviewing the answers in a large group like most days, I tell students to teach their partner how to solve the problems on their initial paper. If there are specific problems that pairs struggled with, I ask a pair to put their work on the board and we solve it as a group. I leave the blank halves in two piles on a table so students can pick up the other half of the worksheet on their way out.

I prefer this activity over games for multiple reasons: it still has the familiar worksheet structure so I don't have to explain any rules; students have to actively participate in their original group to know the material well enough to explain it to a partner; it gives them a chance to verbalize what they are learning – a great way to find out if they truly understand the material; and lastly, it helps the quieter students participate and gain confidence for future sessions.

Comments from students have been positive, even if they initially aren't excited about trying something new. I have noticed that many of the pairs that worked together during this activity have continued to work together through the rest of the semester.

~Addison

Subject Area:

- Math
- Hard Sciences

Grouping:

- Pairs
- Small groups

Notes:

Both halves of the worksheet should be similar in difficulty and require about the same time to complete.

Some students were reluctant to pair up and wasted time, so I just assigned partners for them which worked out fine.

Sharpening Problem-Solving Skills

5.5 Problem-Solving at the Board

Supplies/Prep: Print out and tape up (bring tape!) eight problems around the classroom. This activity works best when there is a lot of board space for students to work.

In my first PAL session of the semester I really wanted to showcase what PAL is all about and how awesome it is! Luckily, a group of 10 chatty students came so we avoided any awkward silence before the session started. I spent a couple minutes introducing PAL, explaining what to expect during sessions, and doing a short icebreaker.

Then I put students into four groups. Each group was told to solve both of the problems on one wall and to write their solutions in a notebook (I had already taped eight problems on four different boards in the room). When they finished with one board, they rotated to the next board, and then the third, until they had completed all the problems. For the last problem set (the last board they got to), I told students to work out the solution on the board itself. So about 10 minutes before the session ended, all the boards were filled with solutions. I then gave them the worksheet with all the problems so they could write down all the solutions before the session ended.

It went great! Students seemed to be genuinely interested in working through the problems and understanding the concepts. The groups worked together really well to solve each problem set. They told me it was helpful to work with someone else because they learned things they didn't know yet or a new shortcut in solving a particular problem.

~ Samara

Subject Area:

- Math
- Hard Sciences

Grouping:

- Small groups

Notes:

Icebreakers are a good way for students to get a feel for the more informal atmosphere of a PAL session.

I ran into some issues with some problems being easier than others so groups got backed up at one of the boards; whenever a group was done early, I asked them to join another group at a different board, even if that group wasn't done.

Because students were on their feet, moving from board to board, I think they felt a little pressure to work at a decent pace.

SECTION 6: Reviewing for Exams

Sections 6 and 7 are both dedicated to activities that will help students prepare for an upcoming exam! The difference is that activities in Section 6 use straight forward study guides and worksheets to review material and Section 7 activities are game-based reviews.

The goal is to give students a chance to assess their skills and make adjustments or changes to their study regimen based on how they do on the in-class worksheet. This set of activities works well for students who don't understand the material as well as they should for the looming exam.

Sometimes just assessing the group dynamics in your session will help you choose which chapter's activities will work best. If your group doesn't like the competitive nature of a game, or they are too stressed to learn the rules of a PAL game, the activities in Section 6 may be a better option.

~Justin

One of the best habits a learner can instill in herself is regular self-quizzing to recalibrate her understanding of what she knows and does not know.

– Peter Brown, *Make it Stick*

Reviewing for Exams

6.1 Lecture Size Sessions

Supplies/Prep: Design a graphic organizer worksheet that suits the course material. Post session: scan notes taken during the session; get students' email addresses to send the compiled lecture summaries – preferably same day, depending on when the exam will be following this activity.

The professor of the course requested that I hold a large (70 students) group review session during her lecture time two days before their first exam. Naturally, I was pretty nervous to be hosting such a large session, but I managed to plan an activity that covered all the material while getting everybody involved. I divided the students into 12 groups (4-5 per group) and assigned each group one of the lectures (there had been 12 lectures). They were given a copy of the graphic organizer to identify, define, and explain terms, important people, and research studies, as well as provide an overall summary of the lecture. The students worked together as a group to review their notes from that lecture and complete the worksheet. At the end of the session, each group gave me their lecture summaries, which I scanned and emailed to everyone.

I heard through the grapevine that the students really liked getting a comprehensive packet of review materials, especially because their professor did not use PowerPoint slides. I was satisfied that nearly every student had been actively involved in the review process.

~Hannah

Subject Area:

- Any

Grouping:

- Small groups

Notes:

The only flaw was that some students were assigned to review a lecture they hadn't attended, meaning they had no notes and couldn't contribute to the summary. I switched those students into a group where they could contribute the lecture notes they did have.

This activity assumes that every student actively takes notes during lecture. Fortunately, I didn't have any students without notes, but this could pose a problem for other facilitators.

6.2 Multiple-Choice Madness

Supplies/Prep: Create a multiple choice exam and print enough copies for each student.

Practice in test-taking skills for college level exams never hurts. For the introductory sociology course I facilitated, my goal was for students to better understand what to expect on their exams. I learned (from the professor and by attending lectures) that the exams would be multiple choice. Using the chapters already covered, I made a practice test for the students to complete in their small groups. To add more of a challenge, I wrote six possible answers (a-f) so that choosing among only four choices on the real exam would seem easier.

This activity was just the right level of difficulty because it gave them practice in both eliminating incorrect answers and making educated guesses. The students worked really well in their groups and seemed eager to apply their knowledge in a test-like format.

~Amber

Subject Area:

- Any

Grouping:

- Small groups

Notes:

Because it was still early in the semester, I let them work in groups to help build relationships among the students and tackle their first “practice exam” together.

For future midterms, I had them work individually, just as they would on a real exam.

Reviewing for Exams

6.3 Reviewing with Sticky Notes

Supplies/Prep: Prepare a set of 12-15 Post-it® notes for each group, with important events, places, or people discussed in lecture and the readings; stack of blank Post-it® notes for each group.

I used this activity as a review and test preparation for my history PAL session. Each of the items on the Post-it notes was associated with the major themes of the course.

At the beginning of the session, I asked the whole group to brainstorm the major themes of the course thus far, which I wrote on the board. Even though students generate most of the activity, I still came to the session with certain themes in mind to use if necessary. For this history course, some of the themes were: Imperialism and Expansion, the 1920s, WWI, the Labor Movement, and Women and Minorities.

I split the students into groups of four and handed each group a stack of fifteen Post-it notes. They were then instructed to put each post-it note under the theme on the board where they thought it best fit. Some of the items connected to more than one theme; in that case I asked them to make copies of the note and place it under all that applied. This aspect encouraged the students to think broadly about the material, further making connections between important items and concepts.

Once the teams had placed all of their notes on the board, we reviewed the themes as a class, with each group explaining in detail why the item fit the theme. This activity was well-received by the students and it allowed us to cover a lot of material in a short amount of time. The explanations are the most important part of this activity as it solidifies the group's understanding of the connection they made, and gives other students a chance to understand and learn that connection.

~Tali

Subject Area:

- Humanities
- Social Sciences

Grouping:

- Small groups

Notes:

I recommend decreasing the number of Post-it notes that each group receives. We ran short on time and had to rush through some of the explanations.

Groups need to be large enough to generate explanations; plus, having fewer groups speeds up the process.

6.4 Who Makes the Study Guide?

Supplies/Prep: Create a review sheet and print one copy per student.

Two weeks before an exam, I like giving my students a lengthy review sheet with pretty specific questions on what their TAs or professor have told them to study. Because it's long, they begin working on it during the PAL session but complete it at home. We go through most of the harder questions towards the end of the session. I encourage them to continue working with each other outside of the classroom. Doing a review this far in advance allows me to ask lots of questions.

An alternative is to have students create parts of the study guide during PAL. Model with an example; make a guide for one chapter/section, distribute them, and give students a limited amount of time to work on the model that you've provided. Then assign each small group a chapter/section to create a similar guide. Collect the student-made study guides, scan them, and send to those who attended the session.

~ Matt

Subject Area:

- Humanities
- Social Sciences
- Hard Sciences

Grouping:

- Pairs

Notes:

I found that much of the learning comes from making the study guide. Doing the alternative suggestion is better for the students. Just have a model that shows the kind of detail expected for the guide to be beneficial.

Reviewing for Exams

6.5 Station to Station

Supplies/Prep: Decide on the number of concepts to review (each station is a specific concept). For each station prepare 2-4 problems, each on a separate worksheet with the solutions on the backside.

During my first semester as a chemistry facilitator, I used a review activity that went really well. I set up six different stations around the room; each station had several problems related to a particular concept, which was labeled above the problems. Working independently or in pairs, students chose which concepts to practice, went to that station, and did the problems (can be either taped to the wall or to the top of the table). I wrote the solutions on the back side of the problem so they could check their answers.

Many students said this was one of their favorite activities because it allowed them to work on only the concepts they needed the most help with. They also liked it because they could work at their own pace and didn't have to wait for others to finish. I found that when the students were able to work at their own pace and choose their own problems, they asked great questions and used each other as a resource. For me, success in a PAL session is witnessing the "ah-ha!" moments, when students finally understand a concept that previously had been difficult for them.

~Jessica

Subject Area:

- Math
- Hard Sciences

Grouping:

- Individual or pairs

Notes:

Because this group was quite large, it was often hard to address every question.

Having an activity where students moved around the room and worked individually allowed them to more easily ask specific questions of me and of each other.

SECTION 7: Game-Based Reviews

The game-based reviews in this chapter can be a welcome change of pace before an exam. While still focused on important concepts, they inject an element of fun! Games are a great choice for a session filled with outgoing students who enjoy competition. A critical factor to their success requires the facilitator to judge the students' level of comprehension of the material. If you feel they are still struggling to master the topics that will be covered on the exam, it's probably better to stick with the review activities in Chapter 6. Game-based reviews work best when the majority of students have already studied the material thoroughly, and instead want more practice in an environment that takes away some of the pre-test stress! Choose games carefully; as Alyssa wisely says in *7.9 Musical Quizzes*, "Sometimes games wrongly reward only the fast thinkers."

~Justin

Research is pretty clear that learning occurs best from games when the games are given a proper introduction, "Here is what you are going to learn when you play this game." ... When thinking about games, think about using them as a method of applying learned knowledge, an opportunity to generate discussion around a topic, as a method of problem-solving and even as a method of thinking about things that normally "won't happen" but are a real distinct possibility.

– Karl Kapp, *Learning games should be part of a learning strategy*

Game-Based Reviews

7.1 Chemistry Bingo

Supplies/Prep: Make a 5x5 grid with a different problem number in each square. (If problems are more difficult, choose a 4x4 grid.) Prepare a list of problems as review for their exam. Each problem corresponds to the numbers on whatever size bingo board you design. Print a separate board for each pair of students – each board should have the numbers organized differently.

Game Rules/Goals: The facilitator designates what type of Bingo formation is needed to win (straight – row or column, “plus” sign, “X”, picture frame).

One creative activity I did last semester was to play Chemistry Bingo! I made a list of problems. I gave each pair a Bingo board with problem numbers randomly written in the boxes. I told them which type of Bingo we would be playing, and gave each pair the list of problems. For example, if it was straight Bingo, a pair could choose which row or column of problems they wanted to solve. Initially, the goal was to be the first team to get Bingo, but they were going through the problems so fast I changed the goal – the group that solved all the problems correctly was the winner. They were really into the game and had a lot of fun. Even those students who usually spaced out got involved.

~Samara

Example boards:

4	3	1	2	22
24	5	25	6	13
20	9	8	14	23
15	7	21	10	17
19	11	16	18	12

Picture Frame

5	10	2	8	9
14	4	12	11	21
7	22	1	24	23
20	15	13	3	17
19	6	16	18	25

Plus Sign

Subject Area:

- Math
- Hard Sciences

Grouping:

- Pairs

Notes:

You might create a set of 5 or 6 boards of each type (5x5 or 4x4), laminate them, and share with other facilitators.

7.2 Bingo with Buildings

Supplies/Prep: Prepare a set of terms/images with corresponding definitions/descriptions – enough for whatever size bingo board you design. Print a separate board for each student – each organized differently, small paper chips (or candy) as markers, copies of answer key for all students.

Game Rules/Goals: Facilitator reads the definitions; students match and cover the corresponding picture or term on their bingo board (see examples below). The first student to complete a row correctly wins.

This was our last PAL session before the final exam in an Introductory Architecture course. For the first half of the session, we played a bingo game using a grid that I had made of images of buildings and diagrams from the textbook. I read a definition, or named the architect, of each building/diagram and students had to cover the appropriate image with a paper chip. At the end, I handed out an answer key with the definitions matched to the correct image.

For the second half, I asked students to form small groups to review old quizzes from the semester. Not all the students had a copy of their past quizzes, but there was at least one copy of each quiz per group. Students could either quiz each other or talk about answers they didn't get right.

Subject Area:

- Humanities
- Hard Sciences

Grouping:

- Small groups

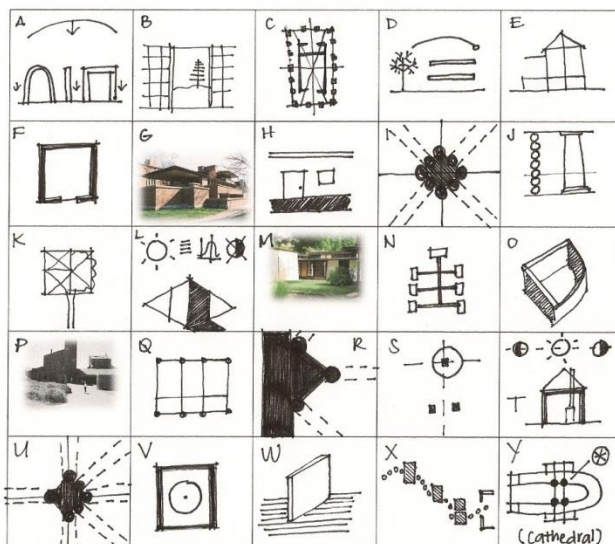
Notes:

If a group had a question, I usually asked another group who knew the answer to explain it.

You might email the students ahead of time to remind them to bring their old quizzes to the session.

~Beth

Bingo board images



Sample Bingo definitions/ID's

- A. Natural Force: Gravity; structures must be able to resist this force and stand
- B. Shoji screens of Katsura.
- C. Parthenon-front
- D. Three conditions of habitation of Villa Savoye:
- E. Tea House of Katsura
- F. Room as Domain, Dogan
- G. Robie House, Frank Lloyd Wright
- H. Openings: reconnect what was separated by walls
- I. Nave column, used in side aisles.
- J. Proportions of a Doric column, 1:7 ratio
- K. Mental and natural order in the world, Katsura
- L. Natural Force: Sunlight, solar radiation
- M. Kings Road House, Rudolph Schindler
- N. Path and Place, Dogan

Game-Based Reviews

7.3 Dueling Teams

Supplies/Prep: Select problems for a worksheet and print two copies. Cut problems into strips, making two piles, with problems in the same order in each pile.

Game Rules/Goals: One student from each team works on the same problem at the board. Each team helps their representative student complete the problem and the first team to finish wins that round.

To prepare for an exam, I usually play some sort of game with the students. I split the group into two teams, each with about 15 people. Two students come up to the board, one from each team. Both students are given the same problem to write on the board for the rest of the students to see. Each student solves the problem, showing their work on the board. They can receive help from their teammates, who should also be working through the problem. Whichever team finishes first receives the point. I then proceed to have the next pair of students come up to the board and do the same. I have received positive feedback about the game, with students saying they like the competition. They also like receiving help from their teammates so they are not doing the problems alone.

~ Kelley

Subject Area:

- Math
- Hard Sciences

Grouping:

- Two teams (or more)

Notes:

Choosing a game for review has its downside; you really need to know and understand the students in your session. Sometimes they are too stressed to play a game right before an exam.

In a session where students attend voluntarily, the competition and emphasis on speed can be counterproductive. Also, they may perceive the game as not serious enough for exam prep.

7.4 Capture the Flag

Supplies/Prep: Decide on the number of stations between flags. For each station prepare 3-4 problems on a worksheet (in case they don't solve the first problem correctly). Prepare answers on backside (prior to the session) so teams may check the solution at the station. Make copies of flags (I used Big Ten college logos).

Game Rules/Goals: Each team of students plays against another team. The goal is to obtain the opposing team's flag and return to your home base before the other team returns to their home base. To advance in the game and obtain the other team's flag, a team must correctly solve one of the problems on the worksheet at each station in order to move to the next station; if their solution is incorrect, they must try the next problem on the worksheet. Once they arrive at the other team's flag, they must answer a problem correctly to obtain the flag. Once the flag is obtained, the team returns to their home base by continuing (counterclockwise) through the stations.

My PAL session played "Capture the Flag". I divided the students into two teams of 3-4 students. For a large group you can have several games going at the same time – just double/triple the materials and have enough space in the room. The layout of the game gave students an opportunity to move around the room, talk to their group members and also see how their teammates solved each problem. It was a nice switch from a regular worksheet.

~Jered

Subject Area:

- Math
- Hard Sciences

Grouping:

- Small groups

Notes:

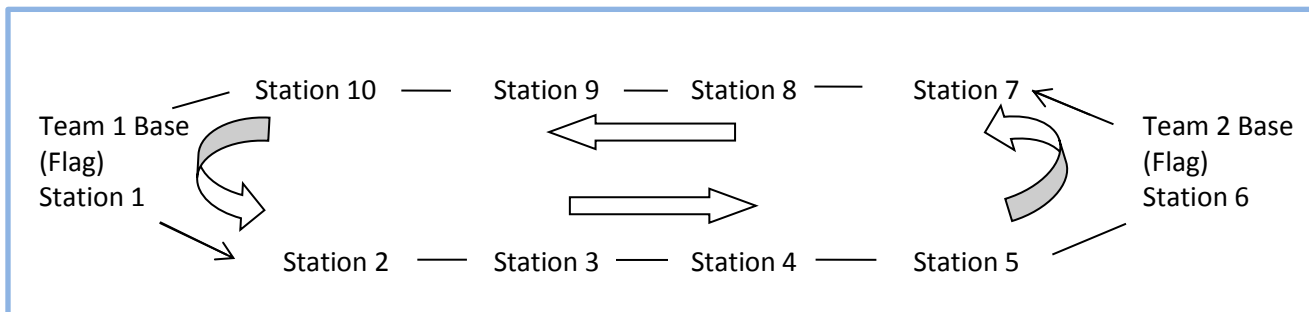
I might have some stations include an individual component, where each group member needs to solve a problem at the station and explain it to their teammates in order for the group to progress to the next station. It could prevent one person dominating the problem-solving process just to win!

Game-Based Reviews

- Arrange the space according to the following diagram: (Example: 10 total stations: 1 for each team's base, 5 per side. Team 1 solves their first problem at Station 1 and proceeds in numeric increasing order, obtains Team 2's flag at Station 6 and returns along the path toward stations 7, 8, etc. Team 1 must solve a "final" problem at Station 1 to win the game.
- Place problems at each station. Distribute flags among teams.
The goal is to obtain the other team's flag and return to home base before the other team.
- To obtain the opposing team's flag, each team must advance through the stations in a counterclockwise path, as in the diagram.
- At each station, there is a set of problems. If the team answers the first problem correctly, they advance to the next station.
- If a team answers the first problem at the station incorrectly, they do the next problem at the same station. If they answer it correctly, they advance to the next station; if not, continue down the list of problems at the same station.
- When a team reaches the other team's base (and flag), they answer the problem at the base in the same manner as the other stations.
- Once the flag is obtained, the team must continue moving counterclockwise.
- Upon arriving at the team's home base, the team correctly answers the final problem at the home base to officially "capture" the other team's flag.

When playing multiple games at the same time, if one game finishes early, other teams could:

- Assist other teams still playing
- Solve other problems from the various stations
- Play again; this time start with problems at the bottom of the list and work up the list



Sample flags (Big 10 Conference)

7.5 Group Relay

Supplies/Prep: Prepare questions from the study guide. Print as many copies as there will be groups. Cut into strips – one pile for each group (each group will get the same set of questions). Students should bring their notebooks and textbooks.

Game Rules/Goals: Like most relay games, one person from the team grabs a question for their team to answer before they can move onto the next question. In this version, peers confirm if the answer is correct. Groups can have only one question at a time, and a different group member should get the next question. The team that completes the most questions in the designated amount of time wins.

One of my favorite activities is a relay race review game I created. The activity works well with groups of 3-4 students and a maximum of four groups. I arrange the groups so that they are in a circle, and then place myself (with the questions) in the center. To start, one member from each group selects a question. The group needs to figure out the answer and ask the group on their left (clockwise) to confirm their answer. If it's correct, the group returns the question to me to exchange it for a new one and to earn their point. It's important to emphasize that the groups must be careful when accepting answers. I remind them that accepting one that doesn't fully or correctly answer the question will negatively impact both groups come test time.

I encourage students to use notes and textbooks. Sometimes I walk around and make sure groups don't stay stuck on one question for too long; however I spend most of the time keeping score and handing out/collecting questions. Depending on the difficulty of the question, students have been able to get through 40 or more questions in 50 minutes. Having other students confirm their answers promotes more relevant discussions than I sometimes hear. It is an easy activity to prepare and has worked well every time I have used it.

Subject Area:

- Any

Grouping:

- Small groups

Notes:

If there are only a few (3-5) students, have them work individually. They still must interact with the other students to confirm their answers.

Another variation is to have the group member who comes up to get a new question explain the answer to the facilitator before getting a new one. Rotating who comes up each time ensures that everyone participates.

~ Summer

Game-Based Reviews

7.6 "X"ed out in Tic-Tac-Toe

Supplies/Prep: Draw a game grid of nine squares on the board. Prepare nine problems (or 18 if you have time to play two rounds). Draw squares to be the same size as the printer paper. Tape the nine premade problems in the squares; cover them with blank printer paper.

Game Rules/Goals: The goal is to connect three X's or O's in a row – horizontally, vertically, or diagonally. A group on the X team selects a question/problem on the grid; if answered correctly, place an "X" in that space. Repeat with the O team; if answered correctly, place an "O." If a group answers incorrectly, any group from the opposing team can answer it and earn an "X" or an "O" on the game grid.

My attempt at Tic-Tac-Toe definitely did not go as well as I had planned because I had a very small voluntary PAL session that day. I think that games in general can be a bad idea when attendance is voluntary because you never know how many students will show up. In this case, only three students attended my session and I did not want to make one student uncomfortable if they had to be on a team by themselves. So, instead of playing the game, the students worked through the problems together. In the end it worked out fine, but in the future I will avoid planning games for my voluntary PAL sessions unless I have very predictable attendance.

~ Jessica

Subject Area:

- Any

Grouping:

- Two teams, small groups within teams

Notes:

Most of the facilitators in our program agree that games aren't as useful with voluntary PAL. We have found it's better to remove the competition and, instead, give a small reward at the end.

With four or more students, this game should work well.

7.7 Jeopardy

Supplies/Prep: Jeopardy template using PowerPoint or similar program; select and organize questions into five different categories; laptop and projector in classroom.

Game Rules/Goals: Groups take turns choosing a category and attempting to answer the question. If correct, group earns the points; if answered incorrectly, question moves counterclockwise to next group, who has a chance to answer it correctly and win the points. Each group takes turns choosing a topic (regardless of who won the previous points). Highest number of points wins.

After four semesters of giving out worksheets at every PAL session, I decided to try something new. In my Calc I PAL session, I decided to model the pre-exam review like a Jeopardy game. I randomly put students into groups of 4 and each group chose a team name. To make sure each group had an equal chance to participate, I had groups take turns choosing and answering the question by moving counterclockwise around the room. If a team got their question wrong, the next team counterclockwise around the room got a chance to answer.

The activity was a rousing success. The students worked hard during the entire session and they enjoyed themselves more than I had anticipated. In the following weeks, I heard indirect testimonials about the effectiveness and entertainment value of the Jeopardy session. While I was sitting in the classroom before my session, students came in saying things like "Playing Jeopardy was so fun! I wish we did more stuff like that in PAL," or "I learned more playing Jeopardy than I do in a week of lectures".

Switching up activities in PAL can be both enjoyable and rewarding for students – I definitely recommend it to anyone with a flair for the unique. Even in sessions where students attend voluntarily and seem more serious, a well-constructed game can work.

~Aaron

Subject Area:

- Any

Grouping:

- Teams

Notes:

Many of the PAL facilitators have attempted Jeopardy, and not always with the same results.

See the next page for more detailed notes on their experiences and recommendations.

Game-Based Reviews

Jeopardy Notes:

- I made one of the five categories be a "fun" category; it was a song identification category, and although it wasn't math-related, I felt it lightened the mood a bit and kept them interested.
- Facilitator Samara found Jeopardy to not work as well with this set of rules. "It was a mandatory chemistry PAL session of 22 students. I set up the game like Aaron, so if a group got a question wrong, it went to the next group. The students in my session were not interested in the game or the problems. If it wasn't their question, they didn't do it, and if one group didn't get a question right, no one else put any effort into solving the problem. I think the main reason this happened was due to the questions I chose. Students either didn't care about the problem because it was too easy and it wasn't their turn, or the problem was too hard to solve anyways, so they didn't bother trying."
- If the game moves too slowly, with the spotlight only on one group at a time, then run the activity more like the real TV game show – all groups try every single problem, and whoever comes up with a solution first is the winner of those points, and gets to choose the next category.
- According to PAL facilitator Lana, "Jeopardy works as a great review for a midterm because it is competitive enough that students are eager to put effort into it, which is something a worksheet doesn't always promote. On the other hand, opting for a study guide over the game may be the right choice for the final review since students could be stressed out and want serious help in studying. The choice often depends on the personality of your group and how close the session is to the actual exam day. Ask them a week ahead so the session best meets their needs."
- "Be wary of using Jeopardy templates with applicable problems already in place, or at least look them over thoroughly beforehand," says Kali. "During my first semester facilitating PAL, I decided to try my hand at Jeopardy to engage the students a bit more. Someone had already made a Jeopardy game for the midterm so instead of coming up with completely new questions, I thought it would be good enough to use the old Jeopardy game (I assumed the questions would be sufficient). During the session, students immediately began choosing the highest points possible (500). Once the 500 point questions were gone, each group went for the lower point questions, but what I hadn't realized was that the problems were built on one another; if you got the 500 point question right, you pretty much automatically got the 400, 300, 200, and 100 point questions right too. This was a huge bust and I was very embarrassed. The moral of the story is, when planning something new, you need to carefully plan out the session and select the questions used for the game – relying on old materials can be dangerous."

7.8 Scratch Off with IF-ATs

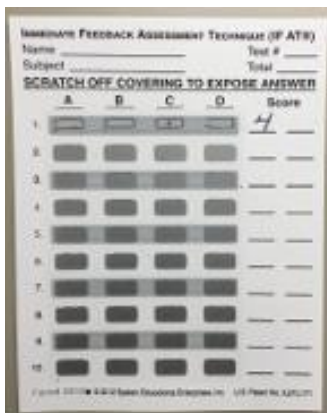
Supplies/Prep: 1 IF-AT (Immediate Feedback Assessment Technique) form per group, 10-question worksheet – 1 sheet per student. IF-AT forms can be ordered at:
<http://www.epsteineducation.com/home/about/>

Game Rules/Goals: Work as a team to scratch off the best answer to each question. If incorrect, proceed to scratch off another choice until answered correctly. Team with the fewest spaces scratched off is the winner (if each question answered correctly, there will be only one space scratched off per question).

I like to make a scratch off practice exam for students in my math PAL session. Although they don't usually have multiple choice tests, it's a fun way to review. They like having multiple choice answers to point them in the right direction for finding an answer, and they like the instant feedback when they scratch off the cards to see if they got it right.

For the first part, I have students work on the test alone, so they can see where their own weaknesses are. Then they get in groups to compare answers or discuss problems they don't know. Then I distribute the IF-AT's, one per group; and for about 10 minutes, they scratch off their answers and try to figure out the ones they got wrong. We spend the rest of the time addressing problem areas or reviewing the most important sections.

~ Alicia



Subject Area:

- Any

Grouping:

- Teams

Notes:

My students enjoy the competition so I award a prize (candy) to the group with the fewest spaces scratched off.

If used as a warm-up activity, you can make a five question worksheet and reuse the IF-AT form at a later session.

Game-Based Reviews

7.9 Musical Quizzes

Supplies/Prep: Device for playing music; one copy of notes from each chapter/lecture; candy (optional).

Game Rules/Goals: Answer other teams' questions correctly to gain points and win the round

An activity I did that went really well was a review game I call, "Musical Quizzes." To prepare for it, I printed out one copy of notes from each lecture that will be covered by the exam. These can be slides that the professor posts, my own notes, or notes provided by the professor. I had students work in pairs but groups of three would work too. Each pair was given a different set of lecture notes and asked to come up with two potential test questions based on that material. I gave them the length of one song, which I played from my computer, to come up with these questions. I typed the questions and emailed them to those who attended so they could review them outside of PAL.

Once the song ended, each team tested the large group with the questions they wrote. Anyone could chime in to answer the question, so it became a bit competitive. Teams earned one point for each correct answer. I kept track of points for each round and the round winner (team that answered the most questions correctly) got to choose either the song for the next round or a piece of candy.

This activity reaches students of all ability and interest levels. Students who have started the studying process can determine what lectures in particular they need to keep reviewing. For students who have not studied yet, this activity gets them to look through the material and pick out the information they think is important. By thinking about potential questions, students practice a valuable study skill – determining and reviewing the most important material.

~Alyssa

Subject Area:

- Any

Grouping:

- Teams
- Pairs

Notes:

Encourage students to choose material that they think will be on the test.

You could have students rotate lecture notes so that, with each round, they generate questions from different chapters/lectures.

If the same students keep chiming in with the answers, be prepared to set an additional rule that will allow everyone time to think before they can respond with the answer. Sometimes games wrongly reward only the fast thinkers.

7.10 Circle of Fortune

Supplies/Prep: Construction paper to create a circle with values (each 100 points or more) similar to Wheel of Fortune (omit "Bankrupt" or "Lose a Turn" spaces). Make spinners from plastic; pins to attach spinner to circle. Prepare 5 problems and their solutions for each value on the wheel (in case a value is landed on repeatedly).

Game Rules/Goals: Answer questions correctly to gain points and win the round.

I created a game for exam review similar to "Wheel of Fortune" but I call it " $(x-h)^2 + (y-k)^2 = r^2$ of Math." (Nerdy, I know.) Divide the group into teams of three people per team. The "wheel" or circle shows values similar to those on the Wheel of Fortune wheel except there aren't spaces for "Bankrupt" or "Lose a Turn" (see example on back). Prepare five problems for each space on the wheel (in the event that a value is landed on repeatedly). One team spins and if they solve the problem correctly, they get the value of the space they landed on. If they do not answer it correctly, they do not get any points, but the other team(s) can steal the question for 50 points (all spaces have a value greater than 100) if they get the right answer, can explain it to the other group, and show them how to correct their error. If there are more than two teams, then the first to get it right gets the 50 points. Turns rotate to the next group (no matter what group gave the correct answer); in this sense it is more like Jeopardy than Wheel of Fortune. At the end, I give out candy prizes to the winning team(s).

~Jered

Subject Area:

- Any

Grouping:

- Teams with small groups within each team.

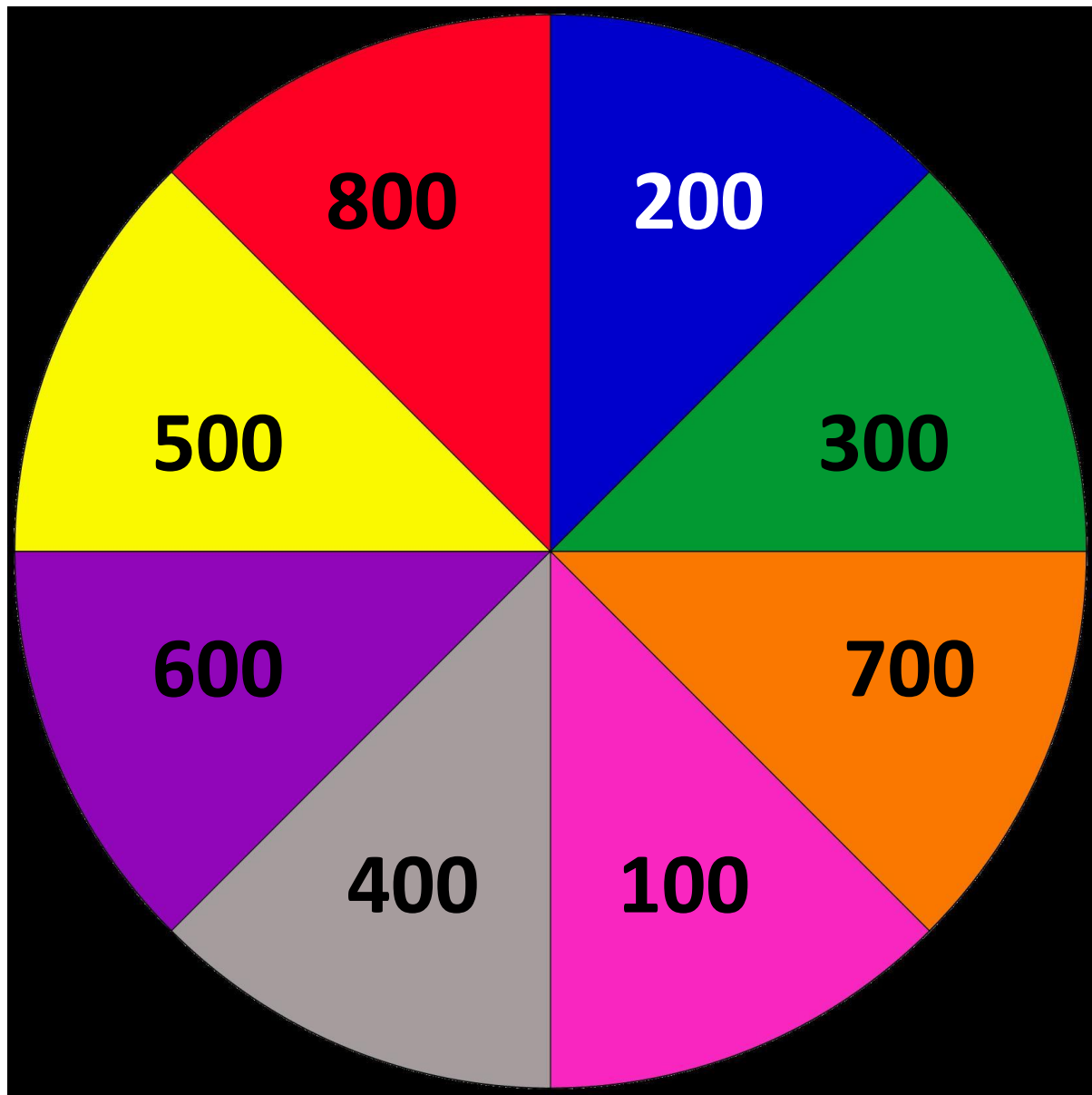
Notes:

Explaining how to correct a team's error is part of the learning.

Laminate the circle board if you plan on using it often.

$$(x - h)^2 + (y - k)^2 = r^2$$

OF MATH



7.11 *Pick Your Present*

Supplies/Prep: Tape, scissors, worksheets. Prep: Gather a supply of magazines and advertisements. Create a worksheet of problems - cut into strips (one problem per strip). Designate a space on the board/wall for each group.

Game Rules/Goals: Students cut out gift ideas from the magazines and ads. A student from each group comes up to get a problem from the facilitator for their group to solve. When a group finishes their problem, facilitator checks the solution; if correct, group gets to tape a gift in their space on the board. There are no winners per se, but the more gifts the better!

As the semester comes to a close in mid-December, I enjoy doing a more light-hearted activity in PAL to ease their stress about final exams. The relaxing part is the first 15 minutes, when students spend time chatting and cutting out gift items in magazines/ads that they would like for the holiday season. After 15 minutes or so, I split the students up into groups and assign each group a spot on the board. Each group then works on one problem at a time, and when completed they verify the answer with me. If correct, the group is allowed to put up one gift in their section of the board. When there are about 10 minutes left in the session, I open it up for students to ask questions about the problems they had the most difficulty with. At the very end, it's fun to have a volunteer from each group come up to the board and explain the gifts their group chose. The gift choices and explanations are often very entertaining – a good way to end the semester.

~Justin

Subject Area:

- Math
- Hard Sciences

Grouping:

- Small groups

Notes:

I usually did this game as the last activity for the semester, but it could be used at any point. I chose to use it at the end because I did not want to give them the expectation that they could have a 15 minute chat-time at the start of every session. If this activity is done at a different time point during the year, just give label it a different label, like “birthday wish list.”

Since this is a more laidback activity, it might not be popular, or wise to use, if their final exam (or any exam) is less than a few days away. It is a good way to get started studying for a final exam or midterm.

Game-Based Reviews

7.12 Quiz Your Peers

Supplies/Prep: Students will need lecture notes and textbook.

Game Rules/Goals: Answer questions correctly to gain points and win the round.

In my PAL session for an intro statistics course, I use an exam prep activity that reviews the material covered in lecture, and helps students identify concepts they don't understand. The last time I used this activity, the students had covered five chapters in lecture since their last exam. I gave everyone five minutes to individually review both their lecture notes and the five chapters in their textbook. Then I grouped the students in pairs. Each pair had 15 minutes to generate four to five test questions in the format that the instructor had used on prior exams, which had been short answer and multiple-choice. In order to get a well-rounded batch of questions, assign each pair a different chapter. This way, students don't all choose questions from the same chapter and neglect to review other material.

For the remaining 30 minutes, each pair asked one question to quiz the next pair, which had a minute to come up with an answer. If they didn't get the correct answer, the question would go to the next pair. One point was awarded to the pair if they answered a question successfully. They were also given a point if they came up with a question that no other group could answer. If none of the groups knew the answer, then the group that came up with the question had to explain the answer to the rest of the class. As the facilitator, I kept track of the time and the points; the winning group got candy.

I have used this activity several times and it always goes well. The students really enjoy it and have given me positive feedback about it. Having students generate their own test questions requires them to consider the most important concepts. The questions also help them to see which areas they need to study on their own and to assess how well they know the material.

~ Cameron

Subject Area:

- Any

Grouping:

- Pairs

Notes:

- If you think you might not get through all the questions, have students write their questions on notecards. Collect the questions at the end of the session, compile them in a document, and email them to the students to use for further review.

7.13 Who Wants to Be a Millionaire?

Supplies/Prep: Make copies of a “Millionaire” worksheet (one per person), with lifeline icons and money/point designations for each question; colored markers – one for each group. Put answers in multiple-choice form. Find at superteachertools.com

Game Rules/Goals: Students select one of four choices to be their final answer for each question. Facilitator gets a copy of groups’ answers written on a blank sheet of paper (students should keep the copy with questions to take home for later review). Problems are reviewed in order on the board, and after each question, the groups are allowed to “walk away” with the amount they have. If a question is answered incorrectly, the group walks away with \$0. The team reaching the highest dollar amount wins the game.

One of the more popular activities I used in Chemistry PAL sessions was the game: “Who wants to be a Millionaire”. The game is a competition to see who can win the most money (each answer is worth an increasing amount of money). Students worked in small groups on the “Millionaire” worksheet. Each group got three lifelines: “phone a friend” was a hint from me; “50/50” was eliminating two of the four possible answers (my choice as to which two); and “ask the book” allowed them to use their book or notes – for that question only. I crossed off each lifeline once it was used; I was actually surprised they didn’t beg for more lifelines. When finished, groups gave me an answer sheet and kept their original worksheet showing work and circled answers. We reviewed the solutions to each problem by having groups come up to the board and show their work. Before we went to the next question, I asked groups if they wanted to end with the money/points they had “won” so far or continue to play (if the group got the problem/question wrong, their total went down to \$0).

This activity is meant for a longer session since it takes time to review each group’s answer individually. Even so, students appreciated doing something fun like this! They were constantly discussing and debating problems throughout the session so I knew it was a productive activity.

~ Justin

Subject Area:

- Any

Grouping:

- Small Groups

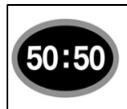
Notes:

One way to shorten the game is to have fewer questions.

It’s best if groups can discreetly alert you that they would like to stop at a certain dollar amount – so as not to influence other groups. If they want to stop and “walk away”, they could lay the colored marker across their worksheet.

Game-Based Reviews

Millionaire Chemistry!



\$100: A balloon is inflated to 4.2 L at a pressure of 1.2 atm at 99°C. What is the final volume if the temperature is decreased to 27°C while the pressure stays constant?

- a. 5.2 L
- b. 3.4 L**
- c. 2.7 L
- d. 1.6 L

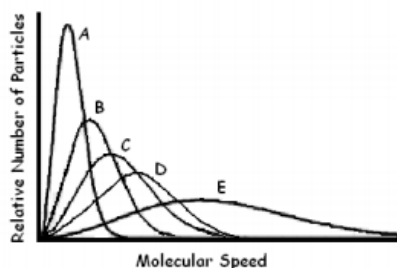
\$200: Which reaction represents a ΔH°_f ?

- a. $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$
- b. $\text{Li}^+(\text{g}) + \text{F}^-(\text{g}) \rightarrow \text{LiF}(\text{s})$
- c. $\text{Al}(\text{s}) + 3/2\text{Cl}_2(\text{g}) \rightarrow \text{AlCl}_3(\text{s})$**
- d. $\text{Pb}^{2+}(\text{aq}) + 2\text{Cl}^-(\text{aq}) \rightarrow \text{PbCl}_2(\text{s})$

\$300: Which of the following processes are exothermic?

- a. The condensation of water on a blade of grass.
- b. The transition of an electron from $n = 3$ to $n = 4$ in a hydrogen atom.
- c. Light emission
- d. More than one of the above are exothermic**

\$500: The following curves represent gases CH_4 , O_2 , He, Ne, F_2 all under identical conditions including temp. Which curve represents Ne?



- a. A
- b. B
- c. C**
- d. D

\$1000: Which of the following objects would you estimate to have the shortest deBroglie wavelength if they are all traveling at the same speed?

- a. Baseball
- b. Marble
- c. Bowling Ball**
- d. Electron

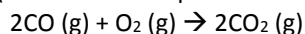
\$2000: Choose the hydrogen atom electron transition below that represents the largest wavelength of photon emitted.

- a. $n=4$ to $n=\text{infinity}$
- b. $n=6$ to $n=5$**
- c. $n=2$ to $n=1$
- d. $n=7$ to $n=1$

\$4000: Determine the frequency (in s^{-1}) of a microwave that has a wavelength of 3.2×10^7 nm.

- a. $5.6 \times 10^{14} \text{ s}^{-1}$
- b. $6.2 \times 10^{-24} \text{ s}^{-1}$
- c. $9.4 \times 10^9 \text{ s}^{-1}$**
- d. $9.6 \times 10^6 \text{ s}^{-1}$

\$8000: What happens to the pressure in a rigid, sealed container as the following reaction proceeds inside (assume the temperature remains constant)?



- a. The total pressure inside the container will decrease.**
- b. The total pressure inside the container will increase.
- c. The total pressure will remain constant.
- d. The pressure will increase dramatically and the container will explode.

\$16,000: Determine the deBroglie wavelength of a tennis ball ($m = 57\text{g}$) traveling 45.0 m/s .

- a. $2.4 \times 10^{-7}\text{ m}$
- b. $2.6 \times 10^{-34}\text{ m}$**
- c. $2.3 \times 10^{-34}\text{ m}$
- d. $1.7 \times 10^{-34}\text{ m}$

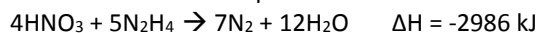
\$32,000: Determine the partial pressure of oxygen in a mixture of argon and oxygen gas if 2.31 g of Ar and 6.22 g of oxygen are placed in a 2 L container at 325K .

- a. 3.36 atm
- b. 2.59 atm**
- c. 1.54 atm
- d. 0.229 atm

\$64,000: Determine the wavelength of light (in nm) that accompanies a transition between the $n=2$ and $n=3$ energy levels of the hydrogen atom.

- a. 434 nm
- b. 821 nm
- c. 365 nm
- d. 657 nm**

\$125,000: Given the following thermochemical equation, determine the mass of water formed when 488 kJ of heat is produced.



- a. 2.95 g
- b. 21.6 g
- c. 35.3 g**
- d. 6.12 g

\$250,000: A 1.922 g sample of C_8H_{18} , is burned in a calorimeter that has a calorimeter constant of 144 J/C and contains 750 g of water. The temperature of the water rose from 14.73 to 42.78°C . Determine the enthalpy of combustion for the C_8H_{18} .

- a. $-5.23 \times 10^3\text{ kJ/mol}$
- b. $-5.01 \times 10^3\text{ kJ/mol}$
- c. -88.0 kJ/mol
- d. $-5.47 \times 10^3\text{ kJ/mol}$**

\$500,000: How many orbitals in an atom can have the $n=4, l=3$ designation?

- a. 3
- b. 5
- c. 7**
- d. 10

\$1,000,000: How many electrons in a single atom can have the $n=3, l=1$ designation?

- a. 1
- b. 2
- c. 4
- d. 6**

SECTION 8: Learning Terms and Vocabulary

Learning to speak the language of any discipline, whether it is chemistry, psychology, or physiology, is critical to succeeding in any course. Section 8 is a collection of activities that help students conquer difficult vocabulary. Without a solid knowledge of the language, exam questions that should be relatively easy are immediately more difficult. These activities offer a variety of methods for students to memorize terms and practice using the vocabulary specific to that course; they are best implemented when a chapter or lecture is vocabulary intensive. These activities also pair well with another type of activity, such as a problem-solving worksheet or graphic organizer that uses the unfamiliar and/or difficult terminology.

~Justin

Direct instruction on words that are critical to new content produces the most powerful learning.

– Robert Marzano, Classroom Instruction that Works: Research-based strategies for increasing student achievement

Learning Terms and Vocabulary

8.1 Blind Key Terms

Supplies/Prep: 1 note card for each key term (I used 21), tape; or sticky notes

This is a great activity when many key terms need to be memorized or reviewed. To prepare, select key words/terms that you want to review with students and write them on a note card or sticky note. I recommend cutting the cards in half so they aren't so big (and you save cards). Make notes of specific points that students need to know about each term so you can prompt them – to make sure they fully understand the key points.

How to use in your session:

1. Tape one key term to each student's forehead (this is where having small note cards helps, because big ones may interfere with their eye sight!). Make sure each student doesn't know what term they have on their head.
2. Have students work together to help each other figure out what term is on their head by explaining key points of that term and/or describing the term. Prompt students as needed if they forget some of the key points.
3. Once a student thinks they know what their term is, let them take it off their head and check. Repeat until all note cards are done.
4. If working in small groups, rotate the cards from group to group so each group covers all the terms.

~ Sara

Subject Area:

- Any

Grouping:

- Small groups

Notes:

Tape one to your head too (for fun), especially if the class is small.

Make sure they use the precise language of the course when explaining or describing the term.

8.2 In Their Own Words

Supplies/Prep: Students will need their books and notes. Make lists of terms – one list for each group.

I facilitate a PAL session for students in an introductory sociology course. The students had just finished reading an important chapter on race issues. I had students form three teams and gave each team a list of terms from the chapter. Without using their books or notes, each group had to create a definition in their own words. I moved around the room during the work time to talk with the groups and check their definitions.

For the second part, students took out their books and notes to expand the definitions, still writing in their own words. To close the session, I asked the students to each write a “one-minute paper” reflecting on what they had learned. I was pleased with their good ideas and comprehension of the vocabulary. I wish the activity had been a little more challenging because the students breezed through it, but I was happy with their reflections in the one-minute papers.

~Amber

Subject Area:

- Any

Grouping:

- Small groups

Notes:

If they get stuck, you can suggest breaking up the word to look for familiar parts.

This activity went more quickly than I had expected, so you might plan an extra activity just in case. You could ask students to explain the terms in context with the main ideas in the chapter.

Have groups share a couple of their most difficult terms with the entire class.

Learning Terms and Vocabulary

8.3 Matching Sentences

Supplies/Prep: Select 12 sentences containing terms, concepts, and themes. Print one copy per group. Cut the sentences separately to make 12 slips; then cut each sentence in half – students will match the first half with the second half.

For one of my PAL sessions, I designed a matching activity to help the architecture students review important terms and concepts. I chose 12 sentences from the chapter that contained important definitions and concepts; I cut each sentence in half so that each small group had a pile of 24 halves to be matched.

As the students worked together to match their halves, I circled around the room and checked the completed sentences. If a group was stuck, they could send a member to another group for help. By doing this, all the groups were able to finish. The students seemed to have fun and actually got competitive, as groups were trying to be first to finish. This activity helped students recognize important points from the reading and understand some of the more difficult vocabulary terms.

~Beth

Subject Area:

- Humanities
- Social Sciences
- Hard Sciences

Grouping:

- Small groups

Notes:

The activity was a bit more challenging than I had anticipated, so I let students use their book and notes to search for hints.

For an added challenge, the two halves, when matched, could create a definition or description of an unnamed term or concept that they then have to identify/name.

8.4 Post-It® Vocabulary

Supplies/Prep: Identify and write applicable terms on sticky notes. One set of sticky notes per group.

For one of my sessions, my goals were for students to have a basic understanding of important terms and events and to know how they fit into the larger course themes. Going through the lectures and the text, I picked out the terms that the professor had emphasized and referenced the most during lecture. I then wrote these terms on individual sticky notes.

During the session, I asked students to generate what they thought the main themes of the course were so far and to write them on the board. I then split the students into teams and gave each team a stack of sticky notes of terms to put under the most appropriate theme they thought it fit. We then discussed each term - why it was placed under the theme and if it fit best there. I found this to be an efficient way to review a fairly large amount of course material.

~ Tali

Subject Area:

- Humanities

Grouping:

- Small groups

Notes:

If you come across a term that the class (and you) believes could be placed in two themes, then have students cut that note in half.

Learning Terms and Vocabulary

8.5 Vocabulary Bingo

Supplies/Prep: Identify at least 25 vocabulary words and create a master list with the corresponding definitions; blank 5x5 grids to use as Bingo boards, one per student (optional, otherwise a blank sheet of paper for everyone).

Game Rules/Goals: Facilitator reads definitions of preselected terms. If a student has the matching term, he/she places a marker over that spot. To win, you need 5 squares marked in a row or column. Winner must explain each term and definition in their winning series.

I write my list of at least 25 vocab words on the board (it helps if you can get to the room early to write the list.) Students create their own 5x5 Bingo cards with spaces for 25 words, writing the words in any order they want on their cards. I number the definitions 1-25 only on my master copy – not the list of terms on the board – and read them one at a time (in numerical order).

When students find the vocab word on their card, they mark it with a small 1 (for the first definition, 2 if it's the second definition, etc.) If a student has a Bingo, they call out the terms they think I read; if they do have Bingo, we go back and discuss each of the five terms. If they do not have bingo, we keep playing. If someone has Bingo, I usually keep going with the same board and the same master list (picking up with the definition where we left off) until one or two more students call Bingo. Once that happens, we start a new round.

After two rounds, or depending on the time, I again read each vocab definition and have the students say what word it is defining. If there is any confusion, we stop and discuss.

~ Samantha

Subject Area:

- Any

Grouping:

- Individual

Notes:

When making Bingo cards, students might select only the terms that are most familiar to them, so to add a little twist, I have them pass their finished card to the person on their left.

To play a second round, create a second master list for yourself with the same definitions only in a very different order. That way, students can use the same board.

Having students write the terms on their grid is good practice in spelling and remembering them.

8.6 Vocabulary Ice-Breaker

Supplies/Prep: Create and print several copies of terms and their definitions (use size 20 font or larger). Cut each sheet so definitions and terms are separated in paper slips, but kept in separate piles; tape for each group to post paper slips to the wall/board.

One of my favorite PAL activities focuses on vocabulary. I have used it in my chemistry and microeconomics PAL sessions (both of which have dense vocab) typically in the first PAL session of the semester, as it's a great combination of icebreaker and learning activity.

To begin, I put all the terms on the left side of the board and the definitions on the right side of the board – in random order. Working in pairs, students go to the board to match terms with their definitions. Each group has a designated spot on the board where they tape their combined terms/definitions. After all the terms have been defined, students can check the other groups' work. I give them time to record the definitions in their notebooks. I really like this activity because it gets students out of their chairs, walking around, and actively talking with each other about the course material. They leave the first PAL session of the semester knowing that we jump right into the course content.

~Deanna

Subject Area:

- Any

Grouping:

- Pairs

Notes:

This activity works best with smaller class sizes, or it can get crowded at the board.

I recommend printing two or three copies of both terms and definitions; tape them on different walls so each group gets to work with all the terms.

Make it even more challenging by including definitions that don't fit any of the terms.

Learning Terms and Vocabulary

8.7 Vocabulary Notecards

Supplies/Prep: 4 sets of notecards, each with 6-10 different vocabulary terms from the chapter.

Students in the introductory journalism course have dozens of terms to learn. A popular activity that I use in my PAL session involves making several lists on the board of terms (6-10 terms per list) so there are anywhere between 25 and 40 terms (depending on the size of the chapter). I group the students into teams, assign each team a list, and hand out an index card for each term. For the first part of the class, students work in their small groups to come up with a definition and an example for each vocab word without using their class notes or textbooks.

After about 10-15 minutes, depending on how discussions are going, I have teams pair up to work together on terms that each team had trouble defining or providing an example. Following this slightly larger group discussion, students have 5-10 minutes to go back and look through their textbook and class notes to complete the remaining definitions on their assigned lists and find an example from the text, which they can compare to their own example. Once they finish, I have each group present their terms to the class, and discuss if there is any confusion.

~ Samantha

Subject Area:

- Any

Grouping:

- Small groups

Notes:

The value of this exercise is that students are creating definitions in their own words and using examples that they are familiar with. Otherwise, anyone could copy a definition out of a textbook.

SECTION 9: On the Shorter Side

Does your planned activity not quite fill the entire time of your session? Are you getting bored with the same routine each session – and so are the students? This section describes activities that you can usually pull off in fifteen minutes or less. Some of them are creative ways to put students in groups at the beginning of the session; others are brain warm-ups, using more in-depth problem solving skills. They can also serve as ways to encourage students to self-monitor their own academic progress. Check out these activities when you are looking to add a little time to your session, or if you want to spice things up and not use only one activity the entire time!

~Justin

To be effective learners, students must adjust their efforts based on their awareness of their own understanding and the level of difficulty of the upcoming task. One of the critical barriers to success for many students may be their inability to objectively assess their mastery of the academic tasks they are facing.

– Randy Isaacson, *Metacognitive Knowledge Monitoring and Self-Regulated Learning: Academic success and reflections on learning*

On the Shorter Side

9.1 Warm-Ups

Lightning

I divide the students into two groups, and give each student a notecard with a number on it: 1 through 8. I then put eight problems on the board; the number on the notecard corresponds to the problem each student needs to solve. When each member of the group completes their problem, one student in the group takes all the notecards with the answers written on them and writes them on the board. The first group to do this wins.

Model: How to present answers

It's worth the extra time early in the semester to explain how you want students to present their solutions to their peers. Emphasize the importance of facing the class, speaking clearly and loudly, and focusing explanations on the thought process. A good presentation says: "why I chose/how I knew to use a particular calculation, chart, or equation," instead of saying "then I multiplied 3 x 75 and got 225."

Grouping/Problem-Solving Combo

Best used when you have predictable attendance. If you want groups of three, make three copies of the worksheet and cut problems into separate strips. Hand out strips randomly. Everyone finds two other people, each with a different problem to form their threesome; each works on his or her question alone (writing solutions in their notebooks) for three minutes, then passes the slip to another member of the group. These are short, not too difficult, review questions.

Self-Select Grouping

Useful when there is a growing division between prepared and unprepared students. To keep the session relevant for all who attend, the facilitator could prepare two different worksheets or activities. Facilitator asks students to write down how they prepared for the week's PAL session. Students then self-select which worksheet/activity to work on. To close the session, ask students to make notes on what they will do to better prepare for the next session. Since making two worksheets is more work for the facilitator, you might do this only a couple of times during the semester as a way to motivate students to reflect on their studying methods.

Reverse Engineering

Give students an equation or solution (pertaining to the week's concepts) and see who can create the most original story/word problem to go with it. Students have an opportunity to connect words with the equation and see their relationship. They often find it harder than they thought.

9.2 Five-Minute Flares

I often find myself getting into a rut with my PAL sessions: warm-up discussion, worksheet, present answers. I also know how hard it can be to continually plan and pull off an extensive activity week after week.

Last semester, I decided to inject a little variety into every session, while still maintaining my general agenda. I started to experiment by creating short five-minute activities and called them "Five-Minute Flares." They were a good way to review certain concepts or work on simpler problems that didn't take long to solve. I also found that they got students talking – about math – right at the beginning of the session.

Most of these activities can be used as either warm-ups or to wrap up a session; some have even served as ways of grouping students. The students liked the variety and it was fun for me to be creative.

~Elaina

Right or Wrong

Supplies/Prep: Slips of paper with correct and incorrect versions of rules or formulas you wish to use.

The first time I tried this was during a session when students were learning their logarithm rules. I took the six properties and printed off a correct and incorrect version of each one. Students got one slip of paper with either the correct or incorrect rule. They had to find their counterpart (i.e. the person with a correct version finds the incorrect), write both versions on the board, and circle the correct rule. The incorrect rule was often one I had seen students mess up, which alerted them to common mistakes to avoid during the exam. This would also work well for exponent laws or definitions.

I had each pair work as partners for the rest of the session so it was an easy way to group everyone.

On the Shorter Side

Match the Graph

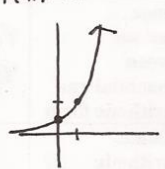
Supplies/Prep: Separate slips of paper – one with a graph, the other with the matching equation (or other topics easily matched).

I distributed the slips of paper (making sure both parts of every pair lands in someone's hands). Students had to find their match (and remained as partners for the rest of the session). It got everyone mixing and talking – about math – right away. I have also made two graphs, one a translation or transformation of the other, and the students had to find each other and then write the equations of both graphs. This was a little more difficult but incredibly helpful in getting students to see how graphs could be manipulated.

Learning Objective Matrix

Supplies/Prep: Prepare an empty grid with the rows and columns already labeled (see example below); one copy per student. Students need their textbooks and notes.

I passed out the grids and gave students five minutes to complete one or two rows in the matrix (using notes and books). When they finished, I checked their matrix – mainly to see they had made a good effort, and then gave them the worksheet. I urged them to finish the matrix on their own because it would be a good study tool. This activity is especially helpful in courses with learning objectives (either given by the professor or listed in the book). You could have them formulate their own learning objectives, which would also be beneficial.

Chapter 5 Review Sheet																
PAL																
Objective	Pages	Summary	Formulas/Properties	Example												
Solve exponential equations	434-441	If bases are equal, use for property or rearrange using other properties	$b^m = b^n$ for $b > 0$ $b \neq 1$	Solve $4^{3x} = 32 \iff$ $2^{2(3x)} = 2^5 \iff$ $6x = 5 \implies \boxed{x = \frac{5}{6}}$												
Graph exponential functions	437-440	domain: all reals examine symmetry test $x=0$ $x=1$ and others	$f(x) = b^x$ $b > 0, b \neq 1$	Graph $f(x) = 2^x$ <table border="1"> <thead> <tr> <th>2^x</th> <th>x</th> </tr> </thead> <tbody> <tr> <td>$\frac{1}{2}$</td> <td>-1</td> </tr> <tr> <td>1</td> <td>0</td> </tr> <tr> <td>2</td> <td>1</td> </tr> <tr> <td>4</td> <td>2</td> </tr> <tr> <td>8</td> <td>3</td> </tr> </tbody> </table> 	2^x	x	$\frac{1}{2}$	-1	1	0	2	1	4	2	8	3
2^x	x															
$\frac{1}{2}$	-1															
1	0															
2	1															
4	2															
8	3															
Solve exponential growth and decay problems	442-451	Includes compound-interest, half-life, and exp functions	Compound Interest $A = P(1+r)^t$ $A = P(1+\frac{r}{n})^{nt}$ $Q = Q_0(\frac{1}{2})^{t/h}$ $A = Pe^{rt}$	\$750 invested 5yrs at 4% compounded semiannually $A = 750(1 + \frac{.04}{2})^{2(5)}$ $\boxed{\$914.25}$												

9.3 Post-Exam Self Evaluation

Supplies/Prep: Copy of Post Exam Self Evaluation printed on both sides (see sample next page). Students should bring copy of the exam – with questions they got wrong, textbook, and class notes. Facilitator can bring notecards (optional).

Many of the freshmen in my chemistry PAL session were able to succeed in high school with little or no studying. Coming to college, where studying is a constant and necessary practice, is a huge change for them. Many of them have admitted that they are overwhelmed by the thought of studying for large exams such as chemistry midterms.

Today I had students bring their copies of the exam they took last week. I prepared a post exam self-evaluation guide for students to complete. This guide has spaces to write down the questions they got wrong, the reason they lost points (didn't understand the question, computation mistake, etc.), and where they could find information on this topic in their textbook or notes. I emphasized that sometimes professors reuse commonly missed questions from the midterms in the final exam. For that reason, it is VERY important that students do not simply give up on a topic they struggled with as soon as the midterm is over. I encourage them to continue this practice after future exams. Doing this will provide students a great way to focus their studies once final exams roll around.

~ Jessica

Subject Area:

- Any

Grouping:

- Individual

Notes:

I usually give students time during the session to create flashcards on difficult concepts from their post-exam self-evaluation.

They find this activity helpful; many admitted that they liked the idea of flashcards but could never find/make the time to sit down and actually do it on their own.

On the Shorter Side

➤ **Handout: Post-Exam Self Evaluation**

CLASS: _____ TEST: _____ CHAPTERS: _____

This worksheet is to help you organize the questions you did not get correct on your past exam. Use this to help identify what area/chapter you may need to brush up on before the final. **Keep this and use it as a resource when you study!**

Question Number	Chosen Answer	Correct Answer	Topic and Chapter
Why did you get this question wrong? a. Guessed/did not know how to do the problem b. Made a computation mistake c. Did not understand the question d. Did not answer the question			
Where could you find review material for this topic? (Textbook page #, Section of notebook, etc.)			

Question Number	Chosen Answer	Correct Answer	Topic and Chapter
Why did you get this question wrong? a. Guessed/did not know how to do the problem b. Made a computation mistake c. Did not understand the question d. Did not answer the question			
Where could you find review material for this topic? (Textbook page #, Section of notebook, etc.)			

Question Number	Chosen Answer	Correct Answer	Topic and Chapter
Why did you get this question wrong? a. Guessed/did not know how to do the problem b. Made a computation mistake c. Did not understand the question d. Did not answer the question			
Where could you find review material for this topic? (Textbook page #, Section of notebook, etc.)			

9.4 Flippy Pages

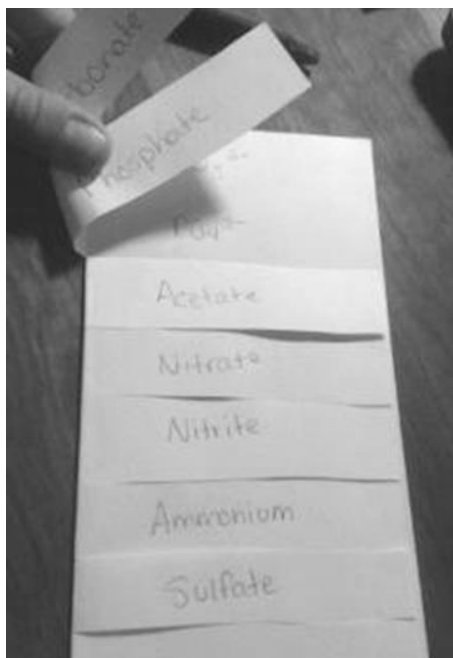
Supplies/Prep: A sample flippy page (see example below); a few pairs of scissors for the session (optional).

Demonstrating study skills doesn't have to be very formal; I often do a quick warm-up/closing activity during the session to show students how I study and why it works for me.

One of my favorite study tools is what I call "flippy pages". They are like flash cards, but in my opinion, much better. Whenever I have to memorize something, like polyatomic ions or functional groups for Organic Chemistry, I use flippy pages. It's like having 15 flash cards all on one page, so I don't worry about losing them the way I might with index cards.

To make a flippy page, take a blank sheet of printing paper and fold it in half the long way. On the top half of the folded sheet, cut flaps that are approximately an inch wide; the outside flap is labeled with the prompt and the inside page reveals the answer. I like to have students make them during PAL so they can use them to study for their exams.

~ Deanna



On the Shorter Side

9.5 Ultimate Exam Page

Supplies/Prep: Students should bring textbook and/or class notes

One of my favorite study skills is what I call the "ultimate exam page." This is simply a sheet of printing paper divided into as many sections as there are tests in the course (can use both sides). Starting with exam 1, students write down ALL the important concepts they need to know for the test, such as things they have to memorize and equations they have to understand. I make time in our session before each exam. They use the same sheet of paper, so when the final exam comes, they essentially have everything they need to review/study on one sheet! I learned this study skill from my previous chemistry PAL facilitator, Samara.

~Deanna

Subject Area:

- Any

Grouping:

- Individual

Notes:

Show a sample page so students can get an idea of how extensive the notes should be. These are just notes regarding what they need to study, not a copy of the entire lecture.

One of the other facilitators distributes 5x8 notecards to accomplish the same goal. She has students take three minutes at the end of the PAL session to add to their cards.

Chem 1050 Exam #1 Study Guide <u>Intermolecular Forces</u> - Ion-Ion - Ion-Dipole - Dipole-Dipole (Hydrogen bonding) - Charge-Induced Dipole - Dipole-Induced Dipole - London (Dispersion) <u>Factors which influence solubility</u> - Lattice Energy - Solvation Energy - Entropy - Like Dissolves Like <u>Mole Fraction</u> $X_1 = \frac{n_1}{n_1 + n_2}$ <u>Wt%</u> $= \frac{wt}{wt + w} \times 100$ <u>Molality</u> $m = \frac{\text{moles solute}}{\text{kg solvent}}$	<u>Vapor Pressure of a Solvent above a Solution</u> (Raoult's Law): equals the mole fraction of the solvent (X_{solvent}) times the vapor pressure of the pure solvent (P°_{solvent}) $P = X_1 P_1^\circ \quad \Delta P = X_2 P_1^\circ$ <u>Boiling Point Elevation</u> - Results because a higher temperature is required to raise the solution's vapor pressure to equal P_{ext} $\Delta T_b = K_b m \quad (T_b - T_b^\circ) = K_b m$ <u>Freezing Point Depression</u> - Occurs because the vapor pressure of the solution is always lower than that of the solvent $\Delta T_f = K_f m \quad (T_f^\circ - T_f) = K_f m$ <u>Osmotic Pressure</u> - Extra hydrostatic pressure - Min pressure needed to stop osmosis - Only solvent passes through semipermeable membrane	$\bar{z}$ = number of ions solute will dissociate into. <u>Henry's Law - Solubility of Gases</u> $S_{\text{gas}} = K_H P_{\text{gas}}$ - Solubility of gas is directly proportional to the partial pressure of the gas above the solution K_H = Henry's Law Constant - Depends on solvent & gas $\frac{S_1}{S_2} = \frac{P_1}{P_2}$
<u>Chapter 16 - Kinetics</u> <u>Reaction Rate</u> - Change in the concentration of reactants (or products) as a function of time <u>Rate Laws</u> - Differential Rate Law - Expression of rxn rate as a function of concentrations - Integral rate law - Expression of concentration as a function of time <u>Reaction Order</u> - 1st order - Rate doubles as [A] doubles $\propto [A]^1$ - 2nd order - Rate quadruples as [A] doubles $\propto [A]^2$ - Zero order - Rate does not change when [A] doubles $\propto [A]^0 = k$	<u>Method of Initial Rates</u> 1) Measure rxn rate over a short time interval @ known initial concentrations 2) Do this several times, varying one initial concentration at a time <u>First Order Reactions</u> - Exponential Decay - If we know k & $[A]_0$ we know $[A]$ @ all times $t_{1/2}$ - time it takes for the reactant concentration to reach 1/2 its initial value - Constant - Independent of reactant conc. $[A] = [A]_0 e^{-kt}$ $t_{1/2} = 0.693/k$ - Radioactive Decay is first order - Carbon Dating <u>Second Order Rxns</u> - Differential $-\frac{\Delta[A]}{\Delta t} = k[A]^2$ $t_{1/2} = \frac{1}{k[A]_0}$ - Integral $\frac{1}{A} = \frac{1}{A_0} + kt$ - Half Life	<u>Reaction Mechanisms</u> - Sequence of single rxn steps that sum to the overall equation - Intermediates = formed and then almost immediately consumed - Elementary processes = individual steps that make up a reaction mechanism; each describes a single molecular event - Relation to Rate Law: rate law for the rate-determining step becomes the rate law for the overall reaction - Rate determining step limits how fast the overall rxn proceeds <u>Collision Model</u>: Two molecules must collide to react - Must have enough energy - Must have correct orientation <u>Activated Energy Diagrams</u> <u>Heterogeneous Reactions</u> <u>Ozone Depletion</u> $\text{NO} + \text{O}_3 \rightarrow \text{NO}_2 + \text{O}_2$ NO catalyzed O₃ destruction

SECTION 10: Why Won't You Tell Us!

Every facilitator in our PAL program has experienced the student(s) who just wants to know the answer or if they are doing the problem right. While facilitators want to be helpful, a hallmark of PAL, and most academic support, is that when a student asks for an answer, the facilitator responds with something other than a simple “correct or incorrect.” Section 10 is different than the previous sections in that these aren't really activities, but rather real accounts of how facilitators handled this situation. They were able to help students understand why their answer was correct or incorrect by using alternative strategies that promoted critical thinking and boosted students' confidence in their problem-solving skills.

~Justin

A major issue for students who attend PAL sessions is understanding the role of the PAL facilitator. The natural belief is that participants ask questions and PAL facilitators provide answers. Changing this dynamic is one of the most challenging and important actions that the PAL Facilitator will take.

– David Arendale, Guide

Why Won't You Tell Us!

10.1 Ask Your Peers First

To do this successfully you need a sense of who in the session seems to grasp the material, and allow that student to support other students.

When students ask me if their work is correct or what the correct answer is, I first delegate the question to other students and to the book for help. As they begrudgingly take out their lecture notes or book, I remind them of what I told them the first day of the PAL sessions – my role is not to teach but to guide them towards the correct answer, and towards feeling confident in their answers. At this point, discussion within the group ensues as they work out what they think might be the correct answer. When I notice a student grasping the concept more quickly than others, I ask him or her to direct the other students to the page number in the book or to paraphrase the book's explanation. This has proven to be very effective because the discussion is engaging and challenges them to learn how to use their book to teach themselves, rather than someone providing the answer.

~ Lia

Notes:

Set the class guidelines and expectations at the beginning of the semester. Let them know that you are not simply going to hand out answers; you will push them to understand it for themselves.

10.2 Confidence is the Issue

The key is to not give in when pressured by students.

One of my chemistry PAL students asked me "Why are we even here?" because I would not confirm if one of her answers was right or wrong. The first question I usually ask them is, "Why aren't you confident in the answer you worked out?" They then stammer a bit and so I ask them what part they are unsure about. Often they say either they just want to know if they are doing it right or they show me a specific step they are unsure of. I usually try to refer them to another group that understands this problem or I ask them leading questions to see if they can get to the solution on their own. If they can show me how they got to this point in the solution, I say, "I would be pretty confident in what you have."

~ Jake

Notes:

Try not to get too frustrated with students who are persistent or rude, as seen in this example. If they see you getting flustered, that's a sign to them that the power has shifted into their hands.

Be cool and calm, and remember that the focus is on learning, which is why you're not flat out providing any answers.

Why Won't You Tell Us!

10.3 No Problem Goes Unsolved

Supplies/Prep: Prepare worksheet of problems. Carry a sheet with the correct answers, written step by step, to all problems on the worksheet.

I am really glad that I emphasized to the students at the beginning of the semester that I am not here just to provide them with answers. When different groups all ask different questions, having the solutions written out step-by-step, while I circulate around the room, helps me spot their mistakes more easily. If students ask for answers, I ask them to walk me through the problem, to show me what they've done so far. If, after they hear their own explanation, it still all makes sense to them, I tell them that they should check their problem-solving steps with their group members.

Students have gotten used to doing this, plus they know we will go over the problems together at the end of the session; they will get their answers eventually, but only after spending time solving the problems. The fact that students know they will eventually get the answers to their questions can be a positive and a negative. Some students see it as an opportunity to keep working towards the correct solution, and others decide to just give up and wait for the answer.

~ Samara

Notes:

Help students avoid giving up by either providing hints or partnering them up with a student who has already completed the question and is willing to demonstrate how they answered it.

10.4 Step by Step

Supplies/Prep: Complete knowledge of the problem – step by step – so you can quickly identify the students' work as correct or incorrect.

This story is basically a duplicate of 10.3, with some additional insight from the facilitator.

When students ask for answers, first I tell them that we will go over the answers together at the end of the session. Next, I have them check with one of their peers. Lastly, if I have time to be absorbed by one group, I tell them I will go over the problem with them. If I do that, we start at the beginning and I have them show me what they've done so far, step by step. If I come across an error, I question that step in a way that hints that something should be altered. If they walk me through the entire problem and all their steps and explanation are correct, I simply say "looks good to me!"

~ Justin

Notes:

Remember to compliment students' work when deserved. Many times, facilitators get caught up in the errors and forget to give support for correct work.

Why Won't You Tell Us!

10.5 Students as Guides

Supplies/Prep: Ability to know who in the session is most proficient in the material, allowing that student to help other students.

When students ask for answers, I generally refer them first to a specific section in the book. Next, I have them ask another group. I have also found that having students float as guides works well when one group completely understands the material and another group is lost and needs help.

Just as the facilitator should not simply hand out answers, neither should the students who are guiding others. You will need to model the difference for students – a good topic for early in the semester.

~ Katie

Notes:

Setting expectations and modeling how you will respond to various situations really need to be established the first few weeks of the semester.

10.6 Value of the Struggle

Supplies/Prep: The following quote (or similar) for presentation to students.

My physics professor offered some valuable insight regarding problem solving, and it is a good story to tell in the first PAL session when you explain you will not be giving them direct answers. As he said, "The problem seems so easy when you have the problem and the solution in front of you, but if a similar problem were placed in front of you, would you be able to answer the question? Probably not, because you still do not know the process behind solving the problem." He made this clear when someone asked him to post the solutions to the practice problems that he had assigned from the book. I didn't believe him until the first exam rolled around; the problems did not seem so easy because I didn't have the answer staring me in the face.

Often the struggle to get to the solution leads to the best learning and best retention of the underlying concepts.

~ Jake

Notes:

Understandably, students will want assurance that answers to the problems will be reviewed at the end of the session.

APPENDIX

PAL Session Plan – sample (developed by facilitators in U of MN PAL program)

SESSION PLAN
Name: Britta
Course: Accounting 2050
Date of Session/Week #: April 14, 2016 – Week #11
Warm Up: 7 minutes
What activity will you use?
We are going to do a similar type of matching activity as last week. I made flashcards again; students will match the sets individually if there are enough or with partners if not. Students will also fill in the blanks of the conceptual problem written on the board - without using notes, if possible.
Grouping Strategy
How will students be grouped?
They will work in partners or groups of three if there is an odd number this week.
Challenging Concepts
What concepts do you expect to be challenging for students this week?
Students will find it difficult to remember the different responses for special circumstances involving bonds sold at par, at a premium, and at a discount; preferred stock and common stock.
Learning Objectives: list up to 3 LO
What will students be able to do by the end of the session that is specific and measurable?
1. Generate journal entries based on an issuance of stock 2. Determine the differences between types of stock and types of bonds 3. Identify financial statement effects of different stock transactions
Activity: 25 minutes
What activity(s) will be used to achieve the LOs?
Students will work on a worksheet that focuses on topics from chapters 10 and 11 this week. Two problems are from the textbook. The other one is more of an all-encompassing problem about different topics. The goal of the worksheet is to provide students with additional practice on topics they have been focusing on the past two weeks.
Presenting/Sharing: 10 minutes
How will students present/share their knowledge discussed during the session?
Students will write answers on the board to one or more of the problems. I will stress that they should write out all of their work so that it is easy for students to see if they did anything differently and then they can ask better questions of the students who are sharing. Each group of students will discuss how they went through each problem at the front of the room.
Recap: 8 minutes
A recap can be done at any time, not just to close the session It can be a wrap-up or a summary — preferably coming from the students – of important material.
Students will have to come up with their own bond problem and find the answer. I may pick a couple of these for the warm-up next week. I'll let students know that next week we will be focusing on topics from chapter 12 and appendix E.

Appendix

Activity Reflection – sample *(developed by facilitators in U of MN PAL program)*

Name of activity Who Wants to Be a Millionaire? **Section** 7.13

Date activity used 4/20 **Material covered** Gas Laws

Grouping strategy used (if applicable) I used playing cards, numbers 2-6. Students picked a card as they walked in. Those with the same numbers became a group.

Any changes made to the original activity? (Offer reasons) I removed the \$300, \$4000, and \$64000 questions to save some time because I didn't know if we would get through all the questions and have time for review.

What worked well about this activity? Because of the competitive nature of the game, the students were focused on the problems the entire time. Also, there was enough time after the game to go over confusing problems. I had students with correct answers put their work on the board and lively discussion followed each question.

What I want to improve next time: I need to do a better job of clarifying the game rules. Next time I would print out the rules and give a copy to each group to read before we start. The explanations took away time and some of the excitement. It was hard to monitor the success of individual groups because they mainly worked on their own (aside from the lifelines). I could have done a better job of walking around and asking groups to explain their thinking to me so I could assess their comprehension.

Did the activity achieve my intended goals? _____

How do I know? What did I observe? Since the exam was 3 days away and most of the students knew the material, it was a fun and successful way to review for the midterm. The group was lively and on task for the entire period. They enjoyed the competition and the relaxed atmosphere reduced some of that pre-exam stress.

I give this activity/session _____ stars ☆ ☆ ☆ ☆ ☆

The above guide can be used to reflect on your own peer learning activity, whether it is based on one from this collection or one of your own. Either way, reflection is an important part of facilitation. Consider both the positive and negative aspects of the session to determine if the students achieved the goals you had in mind. Push yourself to think critically about what worked and what didn't. Just because the students completed the activity doesn't mean they truly saw improvement in their learning. Analyze what they said as well as the peer-to-peer interaction: was there too much down time or not enough time? What were the group dynamics? How about the attitudes of individual students? Did some groups fall behind? This reflection can be an excellent resource for shaping your future sessions; plus it will be valuable to review if you or other facilitators want to use the activity another time. Best of luck!

~Justin

➤ *Session Planning Form*

SESSION PLAN Name: _____ Course: _____ Date of Session/Week #: _____
Warm Up: _____ minutes What activity will you use?
Grouping Strategy How will students be grouped?
Challenging Concepts What concepts do you expect to be challenging for students this week?
Learning Objectives: list up to 3 LO What will students be able to do by the end of the session that is specific and measurable? 1. 2. 3.
Activity: _____ minutes What activity(s) will be used to achieve the LOs?
Presenting/Sharing: _____ minutes How will students present/share their knowledge discussed during the session?
Recap: _____ minutes A recap can be done at any time, not just to close the session It can be a wrap-up or a summary — preferably coming from the students – of important material.

Attach a copy of your worksheet or main handout

Appendix

➤ *Activity Reflection Form*

Name of Activity: _____

Date used: _____ Material covered: _____

Grouping strategy used (if applicable): _____

Any changes made to the original activity? (Offer reasons) _____

What worked well about this activity? _____

What I want to improve next time: _____

Did the activity achieve my intended goals? _____

How do I know? What did I observe? _____

I give this activity/session _____ stars



➤ *Session Planning Form*

SESSION PLAN Name: _____ Course: _____ Date of Session/Week #: _____
Warm Up: _____ minutes What activity will you use?
Grouping Strategy How will students be grouped?
Challenging Concepts What concepts do you expect to be challenging for students this week?
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Recap: _____ minutes A recap can be done at any time, not just to close the session It can be a wrap-up or a summary — preferably coming from the students – of important material.

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Appendix

➤ *Activity Reflection Form*

Name of Activity: _____

Date used: _____ Material covered: _____

Grouping strategy used (if applicable): _____

Any changes made to the original activity? (Offer reasons) _____

What worked well about this activity? _____

What I want to improve next time: _____

Did the activity achieve my intended goals? _____

How do I know? What did I observe? _____

I give this activity/session _____ stars ☆ ☆ ☆ ☆ ☆

➤ *Session Planning Form*

SESSION PLAN	
Name: _____	
Course: _____	
Date of Session/Week #: _____	
Warm Up: _____ minutes	
What activity will you use?	
Grouping Strategy	
How will students be grouped?	
Challenging Concepts	
What concepts do you expect to be challenging for students this week?	
Learning Objectives: list up to 3 LO	
What will students be able to do by the end of the session that is specific and measurable?	
1.	
2.	
3.	
Activity: _____ minutes	
What activity(s) will be used to achieve the LOs?	
Presenting/Sharing: _____ minutes	
How will students present/share their knowledge discussed during the session?	
Recap: _____ minutes	
A recap can be done at any time, not just to close the session It can be a wrap-up or a summary — preferably coming from the students – of important material.	

Attach a copy of your worksheet or main handout

Appendix

➤ *Activity Reflection Form*

Name of Activity: _____

Date used: _____ Material covered: _____

Grouping strategy used (if applicable): _____

Any changes made to the original activity? (Offer reasons) _____

What worked well about this activity? _____

What I want to improve next time: _____

Did the activity achieve my intended goals? _____

How do I know? What did I observe? _____

I give this activity/session _____ stars ☆ ☆ ☆ ☆ ☆

Endnotes:

Preface:

1. The eight principles of PAL, David Arendale & Mary Lilly, *Guide for Peer Assisted Learning (PAL) Group Facilitators* (2012), 9-10. <http://z.umn.edu/facilitatortoolkit>

1. Applying Study Skills:

Epigraph: Graham Gibbs, *Times Higher Education*, May 30, 2013, retrieved from: <http://www.timeshighereducation.co.uk/news/raising-awareness-of-best-practice-pedagogy/2004204.article>

2. Grasping Concepts:

Bloom's Taxonomy image from University of North Carolina. Retrieved July 3, 2014 from: <http://www.unc.edu/learnnc/kinetic-connect/noframes.html>

3. Promoting Participation:

Epigraph: Greg Light and Marina Micari. *Making Scientists: Six principles for effective college teaching* (Cambridge: Harvard University Press, 2013), 104.

4. Solidifying Material:

Epigraph: Peter Brown, Henry Roediger III, Mark McDaniel, *Make it Stick: The science of successful learning* (Cambridge: Harvard University Press, 2014), 6.

5. Sharpening Problem-Solving Skills:

Epigraph: Light, *Making Scientists*, 77.

6. Reviewing for Tests:

Epigraph: Brown, *Make it Stick*, 21.

7. Game Based Reviews:

Epigraph: Karl M. Kapp, retrieved June 10, 2014 from: <http://karlkapp.com/learning-games-should-be-part-of-a-learning-strategy/>

8. Learning Terms and Vocabulary:

Epigraph: Robert Marzano, Debra Pickering, Jane Pollack, *Classroom Instruction that Works: Research-based strategies for increasing student achievement* (Alexandria: ASCD, 2001), 127.

9. On the Shorter Side:

Epigraph: Randy Isaacson, "Metacognitive Knowledge Monitoring and Self-Regulated Learning: Academic success and reflections on learning," *Journal of the Scholarship of Teaching and Learning*, Vol. 6, No. 1, August 2006, 39.

10. Why Won't You Tell Us!:

Epigraph: Arendale, *Guide*, 78.

Index

INDEX

Board Work	1.3, 1.4, 1.8, 1.9, 2.6, 3.3, 4.2, 4.4, 5.1, 5.3, 5.4, 5.5, 6.3, 7.3, 7.13, 8.4, 8.6, 8.7, 9.1, 9.2
Graphic Organizers:	
charts	2.1, 3.5, 4.2, 6.1
concept maps.....	1.3
outlines	1.3, 1.5, 1.6, 1.7, 1.11
timelines	1.9
Venn diagrams	1.8, 4.2
Grouping:	
individual.....	1.11, 3.3, 3.7, 6.5, 8.5, 9.1, 9.3, 9.5
large discussion.....	1.1, 1.2, 2.6, 3.4
pairs	1.3, 1.7, 1.10, 2.1, 2.5, 3.1, 4.3, 5.2, 5.4, 6.4, 6.5, 7.1, 7.9, 7.12, 8.6
small groups.....	1.1, 1.2, 1.4, 1.5, 1.6, 1.8, 1.9, 2.2, 2.3, 2.4, 3.2, 4.1, 4.2, 4.4, 4.5, 5.1, 5.3, 5.4, 5.5, 6.1, 6.2, 6.3, 7.2, 7.4, 7.5, 7.10, 7.11, 7.13, 8.1, 8.3, 8.4, 8.7
teams	4.4, 7.1, 7.3, 7.4, 7.5, 7.6, 7.7, 7.8, 7.9, 7.10, 7.13, 8.2, 8.4, 8.7
Notecards	1.4, 3.7, 8.7, 9.1, 9.3, 9.4
PowerPoint	1.11, 3.6, 5.3, 7.7
Problem-solving . 2.2, 3.2, 3.3, 3.7, 4.4, 4.5, 5.1, 5.3, 5.4, 5.5, 6.5, 7.1, 7.3, 7.4, 7.6 7.7, 7.8, 7.10, 7.11, 7.13, 9.1,	87, 10.3, 10.6
Station Activities	3.7, 5.1, 5.2, 6.5, 7.4
Study Skills:	
active reading	1.2, 1.3
concept identification	1.3, 1.4, 1.8, 2.1, 2.3, 2.5, 3.6, 3.7, 5.5, 6.3, 6.5, 7.12, 8.3, 9.1, 9.2, 9.5
definitions	2.1, 2.6, 7.2, 8.2, 8.3, 8.4, 8.5, 8.6, 8.7
graphs	2.2, 9.2
note-taking.....	1.2, 1.5, 1.6
outlining.....	1.3, 1.4, 1.5, 1.6, 1.7, 1.11
self-monitoring	1.11, 3.1, 3.7, 9.3
study guides	1.11, 3.6, 6.1, 6.4, 7.5, 7.7, 9.2, 9.4, 9.5
theme identification	1.3, 2.6, 4.2, 4.3, 6.3, 8.4
Subject:	
any.....	1.3, 1.5, 1.6, 1.7, 1.11, 3.1, 3.3, 4.4, 6.1, 6.2, 7.5, 7.6, 7.7, 7.8, 7.9, 7.10, 7.12, 7.13, 8.5, 8.6, 8.7, 9.3, 9.5
hard sciences	1.4, 2.1, 2.2, 2.5, 3.7, 5.1, 5.2, 5.3, 5.4, 5.5, 6.4, 6.5, 7.1, 7.2, 7.3, 7.4, 7.11, 8.3
humanities	1.1, 1.2, 1.4, 1.8, 1.9, 1.10, 2.3, 2.4, 2.6, 3.2, 3.4, 3.5, 3.6, 4.1, 4.2, 4.3, 6.3, 6.4, 7.2, 8.1, 8.3, 8.4
math.....	3.7, 5.1, 5.2, 5.3, 5.4, 5.5, 6.5, 7.1, 7.3, 7.4, 7.10, 7.11
social sciences....	1.1, 1.2, 1.4, 1.8, 1.9, 1.10, 2.1, 2.3, 2.4, 2.6, 3.2, 3.4, 3.5, 3.6, 4.1, 4.2, 4.3, 4.5, 6.3, 6.4, 8.3
Test Preparation	1.4, 1.11, 3.5, 4.4, 6.1, 6.2, 6.3, 6.4, 6.5, 7.1, 7.2, 7.3, 7.5, 7.7, 7.8, 7.9, 7.10, 7.12, 7.13, 8.1, 8.3, 8.4, 9.1, 9.2, 9.3, 9.4, 9.5, 10.6
Worksheets ...	1.7, 1.10, 1.11, 2.2, 3.2, 3.5, 4.5, 5.3, 5.4, 5.5, 6.1, 6.5, 7.3, 7.4, 7.7, 7.8, 7.11, 7.13, 9.1, 9.2, 10.3
Writing	1.6, 1.10, 1.11, 2.4, 3.4, 4.1, 4.2, 4.3
thesis.....	1.1, 1.10